

SECTION 230500 - COMMON WORK RESULTS FOR HVAC

PART 1 - GENERAL

1.1 Scope of Work

- A. The Contractor shall provide all materials, equipment and labor necessary to install and set into operation a complete mechanical systems as shown on the engineering drawings and as specified herein.

1.2 Quality Assurance

- A. See the General and Supplementary General Conditions.
- B. All work shall be in accordance with State Code and Underwriter's Regulations. Minimum requirements shall be the State Plumbing, Mechanical, Gas, and Energy Code.
- C. Wherever the words "Approved", "Approval", or "Approved Equal" appear, it is intended that items other than the model numbers specified shall be subject to the approval of the Engineer.
- D. "Provide" as used herein shall mean that the Contractor responsible shall furnish and install said item or equipment. "Furnish" as used herein shall mean that the Contractor responsible shall acquire and make available said item or equipment and that installation shall be by others. "Install" as used herein shall mean that the Contractor responsible shall make installation of items or equipment furnished by others.
- E. All material and equipment that the Contractor proposes to substitute in lieu of those specified, shall be submitted to the Engineer ten (10) days before the bid date for evaluation. The submittal shall include a full description of the material or equipment and all pertinent engineering data required to substantiate the equality of the proposed item to that specified. Items that are submitted for approval after this date will not be accepted. The General Conditions will be followed for substitutions after award of the contract.

1.3 Submittals

- A. See General and Supplementary General Conditions.
- B. Within twenty days after notification of the award of the Contract and written notice to begin work, the Contractor shall submit to the Architect/Engineer for approval a detailed list of equipment and material which he proposes to use. Items requiring submittal data for approval will be noted at this time. Six (6) sets of submittal data shall be provided for approval
- C. Each submittal shall bear the approval of the Contractor indicating that he has reviewed the data and found it to meet the requirements of the specifications as well as space limitations and other project conditions. The submittals shall be clearly identified showing project name, manufacturer's catalog number, and all necessary performance and fabrication data. Detailed submittal data shall be provided when items are to be considered as substitutions for specified items. Acceptance for approval shall be in writing from the Engineer.

1.4 Product Delivery, Storage and Handling

- A. All material and equipment shall be delivered and unloaded by the Contractor within the project site as noted herein or as directed by the Owner.
- B. The Contractor shall protect all material and equipment from breakage, theft, or weather damage. No material or equipment shall be stored on the ground.
- C. The material and equipment shall remain the property of the Contractor until the project has been completed and turned over to the Owner.

1.5 Work Conditions and Coordination

- A. The Contractor shall review the electrical plans to establish points of connection and the extent of electrical work to be provided in his Contract. All electrical work shall be performed by a licensed electrician.
- B. Electrical work shall be in accordance with State codes, and as specified in Division 16 contained herein.
- C. Pipe chases required for installation of work shall be provided by the General Contractor unless otherwise noted. This Contractor shall be responsible for coordinating the location of all required chases.
- D. All work shall be coordinated with other trades. Cutting of new work and subsequent patching shall be at the Contractor's expense at no extra cost to the Owner.

1.6 Guarantee

- A. Where items of equipment or material carry a manufacturer's warranty for any period in excess of twelve (12) months, then the manufacturer's warranty shall apply for that particular piece of equipment or material. The contractor shall replace such defective equipment or materials, without cost to the owner, within the manufacturers warranty period.
- B. The contractor shall unconditionally guarantee materials and workmanship against patent defects arising from faulty materials, faulty workmanship or negligence for a period of twelve (12) months following the final acceptance of the work and shall replace such defective materials or workmanship without cost to the owner.
- C. The contractor shall provide a five year compressor warranty for all refrigeration compressors from date of system acceptance.

PART 2 - PRODUCT

- 2.1 Materials and equipment shall be new, unless noted otherwise, of the highest grade and quality and free from defects or other imperfections. Material and equipment found defective shall be removed and replaced at the Contractor's expense.
- 2.2 The Contractor shall provide nameplates for identification of all equipment, switches, panels, etc. The nameplates shall be laminated phenolic plastic, black front and back with white core, white engraved letters (1/4" minimum) etched into the white core. Nameplates shall be fastened with pan head tapping screws.

PART 3 - EXECUTION

3.1 Inspection

- A. This Contractor shall examine the areas of completed work and shall insure that no defects or errors are present which would result in the poor application or installation of subsequent work.

3.2 Installation

- A. All work shall be performed in a manner indicating proficiency in the trade.
- B. All pipes shall be either parallel to building walls or plumb where installed in a vertical position and shall be concealed when located in architecturally finished areas.
- C. Any cutting or patching required for installation of this Contractor's work shall be kept to a minimum. Written approval shall be required by the Architect/Engineer if cutting of primary structure is involved.
- D. All finishing shall be by the General Contractor.
- E. The Contractor shall lay out and install his work in advance of pouring concrete floors or walls. He shall furnish all sleeves to the General Contractor for openings through poured masonry floors or walls, above grade, required for passage of all pipes required to support his equipment.
- F. All fixtures shall be accurately roughed in according to the manufacturer's installation dimensions so that no offset adaptors, flexible connections or other improvising are necessary. All incorrect work shall be torn out and corrected and walls and floors patched.

3.3 Performance

- A. The Contractor shall perform all excavation and backfill operations necessary for installation of his work.
- B. Rock excavation shall be defined in the Supplementary General Conditions. Unless specifically stated, neither rock excavation nor a unit price for rock excavation shall be required in the bid.

3.4 Erection

- A. All support steel, angles, channels, pipes or structural steel stands and anchoring devices that may be required to rigidly support or anchor material and equipment shall be provided by this Contractor.

3.5 Adjust and Clean

- A. All equipment and installed materials shall be thoroughly clean and free of all dirt, oil, grit, grease, etc.

- B. Factory painted equipment shall not be repainted unless damaged areas exist. These areas shall be touched up with a material suitable for intended service. In no event shall nameplates be painted.
- C. At a scheduled meeting, the Contractor shall instruct the Owner or the Owner's representative in the operation and maintenance of all equipment installed under his Contract.

3.6 Maintenance and Operating Manual

- A. The Contractor shall prepare four (4) copies of a manual describing the proper maintenance and system operation. This manual shall not consist of standard factory printed data intended for dimension or design purposes (although these may be included), but shall be prepared to describe this particular job. This manual shall include the following:
 - 1. Index and page numbers.
 - 2. Certificate of substantial completion.
 - 3. A summary sheet of warranties with the dates noted and a copy of all warranties.
 - 4. List of all subcontractors and suppliers with names, addresses and phone numbers.
 - 5. Certified testing and balancing report.
 - 6. All submittal data and shop drawings.
- B. The O & M manuals shall be installed in 3 ring heavy back note books with the name of the building and the words, "Operations and Maintenance Manuals" permanently affixed to the cover and spine.
- C. The operating and maintenance manuals shall be submitted to the Engineer (2) weeks before the pre-final inspection, for approval. When the manuals are considered complete by the Engineer, they will be turned over to the Owner for their permanent use.

3.7 As Built Documents

- A. The Contractor shall submit to the Engineer a set of accurately marked-up plans indicating all changes encountered during the construction. Final payment will be contingent upon receipt of these as-built plans.

3.8 Training

- A. The contractor shall provide owner training in not less than two sessions on the operation and maintenance of the systems installed by this contract
- B. Sessions shall be digitally recorded and turned over to owner
- C. Contractor shall provide record of individuals trained.

END OF SECTION 230500

SECTION 230513 - ELECTRICAL WORK (MECHANICAL)

PART 1 - GENERAL

- 1.1 This Contractor shall be responsible for the entire control system and control connections to all equipment installed as part of his contract.
- 1.2 Wiring from disconnect switches, junction boxes, etc. up to mechanical equipment shall be by this contractor. Final electrical connections to mechanical equipment shall be by this contractor.
- 1.3 All power and control wiring shall be in conduits.
- 1.4 All electrical work shall be performed by a licensed electrician.
- 1.5 All electrical work shall be in accordance with the State Building Code and all its supplements and the latest edition of the National Electrical Code.

PART 2 - PRODUCT

- 2.1 All motor starters, disconnects, switches, relays, conduits, conductors, etc. that are required for a complete electrical power and/or control system shall conform to the requirements set forth by NEC.
- 2.2 Refer to the plans for the type, size and electrical characteristics of the starters, disconnects, switches, relays, conductor and conduits.
- 2.3 All conductors and conduits shall be sized as noted on the plans or as required per NEC.

PART 3 - EXECUTION

- 3.1 All motor starters, disconnects, and switches shall be installed on or as close to the equipment they are serving as possible, or where shown on the plans.
- 3.2 Electrical connection to equipment subject to vibration which develops objectionable noises shall be made from the conduit system with short lengths of flexible "Liquid- Tite" conduit. Connection to other equipment shall be made with rigid conduit.
- 3.3 Conduits shall be run in a concealed space such as wall cavities, ceiling cavities, etc. except in the mechanical rooms where conduit may be run exposed.

END OF SECTION 230513

SECTION 230529 - MECHANICAL HANGERS AND SUPPORTS

PART 1 - GENERAL

- 1.1 This Section includes all hangers and supports, etc. as may be required to provide a complete piping system.
- 1.2 The actual arrangement of the piping shall follow the general locations shown on the Drawings, such that clearances, line drainage, etc. shall be maintained.
- 1.3 Refer to specification Section 232113 for piping.

PART 2 - PRODUCT

- 2.1 Piping shall be as stated in Piping Section(s).
- 2.2 Hangers and supports shall be as manufactured by B-Line Systems, Inc., PHD Manufacturing, Empire, or Modern Support Devices.

PART 3 - EXECUTION

- 3.1 In no case shall this Contractor be allowed to cut or reduce the specified insulation covering to allow the application of a smaller hanger than required.
- 3.2 Hangers shall be spaced as dictated by North Carolina Mechanical Code.
- 3.3 Hangers shall be provided at each change in direction.
- 3.4 Vertical risers shall be supported at each floor, 5 feet on center, and/or at changes in direction of pipe.
- 3.5 Do not support piping from bar joist bridging and/or roof deck.
- 3.6 Supports shall have rubber insert or painted surface to prevent dissimilar metals from direct contact.
- 3.7 Contractor shall provide threaded rod attachment to structure as required for load at support.
- 3.8 Only plumbing work shall be supported by hangers provided by this contractor.

END OF SECTION 230529

SECTION 230553 - IDENTIFICATION OF HVAC COMPONENTS

PART 1 - GENERAL

- 1.1 This section includes insulation for piping, ductwork, and equipment, as shown on the plans.
- 1.2 All coverings, and adhesives shall have a flame spread classification of 25 or less and a smoke developed rating of not more than 50.

PART 2 - PRODUCT

PART 3 - EXECUTION

- 3.1 Equipment
 - A. All HVAC equipment, including air handlers, fans and pumps shall be properly identified with equipment identification, equipment controlled, electrical ratings and date of installation.
 - B. Equipment shall be clearly identified with engraved phenolic plates securely fastened to the equipment with sheet metal screws. Phenolic plates shall be white background and black lettering.
 - C. All serviceable equipment (fans, reheat coils, VAV boxes, etc.) located above ceilings or other concealed spaces shall clearly identified on an adjacent finished surface below service space. Label shall be engraved phenolic plate with white background and white letters. Label shall list name of equipment.
 - D. Equipment labeling shall be coordinated with owner to match identification used by Building Automaton System.

- 3.2 Ductwork

- A. Paint all exposed ductwork insulation in mechanical rooms white. Ductwork exposed in finished spaces shall be painted as shown on architectural plans.

Duct System	Color Stencil Identification	Label Color	Lettering Color
Supply Ductwork	SUPPLY AIR	Green	White
Return Ductwork	RETURN AIR	Blue	White
Relief Ductwork	RELIEF AIR	Blue	White
OA Ductwork	OUTSIDE AIR	Blue	White
Exhaust Ductwork	EXHAUST	Yellow	Black

- 3.3 Piping and Valves

- A. Valve Identification

1. All valves shall be tagged brass valve tags with chains for isolation and control valves.
 2. Provide valve tag chart in the O&M manual.
 3. Provide famed valve tag chart with lexan cover mounted in each mechanical room. Chart shall include all valves in that room.
 4. Include the tag numbers in the as-built drawings.
- B. All piping shall be provided with identification in accordance with ANSI A13.1-1981 standards. Markers shall be located at each wall, floor or ceiling penetration, and at every 20 ft. Markets shall be fully legible from floor level showing medium contained pipe, and direction flow. Stenciling as indicated below will be acceptable in lieu of markers.
- C. All exposed piping in mechanical rooms shall be painted and marked as listed below:
- | Piping System | Color | Stencil Identification | Label Color | Lettering Color |
|-------------------|-------------|------------------------|-------------|-----------------|
| Condensate Return | Gray | COND | Green | White |
| Chilled Water | Dark Blue | CHW | Green | White |
| Heating Water | Dark Orange | HHW | Green | White |
| Natural Gas | Yellow | GAS | Yellow | Black |
- D. Pipe identification shall contrast in color to the pipe colors and be easily readable. The width of color bands should be equal to the size of the stencil indicated below.
- E. For insulated pipe systems, stencil sizes are as follows:
1. For pipes up to 1 inch, use 1/2 inch letters.
 2. For pipes 1 inch to 2 inches, use 3/4 inch letters.
 3. For pipes 2 inches to 4 inches, use 1 1/4 inch letters
 4. For pipes 4 inches to 6 inches, use 1 1/4 inch letters.
- F. f. For un-insulated systems, stencil sizes are as follows:
1. For pipe diameters up to 1 inch, use 1/2 inch letters.
 2. For pipe diameters from 1 inch to 2 inches, use 1 inch letters.
 3. For pipe diameters from 2 inches to 6 inches, use 2 inch letters.

END OF SECTION 230553

SECTION 230593 - TESTING AND BALANCING

PART 1 - GENERAL

1.1 Section Includes

A. Testing, Adjusting, and Balancing:

1. Air condition equipment, including air distribution devices, supply ducts, air handling units, condensing units, fans, coils, and related equipment.
2. Hydronic systems, including pumps, water distribution systems, chillers, boilers, heat exchangers, coils, and related equipment.

1.2 References

A. American Society of Heating, Refrigerating and Air-conditioning Engineers (ASHRAE)

1. Standard 111-2008 – Measurement, Testing, Adjusting and Balancing of Building Heating, Ventilation, Air-conditioning and Refrigeration Systems.
2. Applications Handbook 2019, Chapter 39 – Testing, Adjusting, and Balancing

B. Testing, Adjusting and Balancing Bureau (TABB) – International Standards for Environmental Systems Balance.

C. Sheet Metal and Air Conditioning Contractors' National Standards for Total System Balance.

D. Associated Air Balance Council (AABC) – National Standards for Total System Balance.

E. National Environmental Balancing Bureau (NEBB) – Procedural Standards for Testing, Adjusting and Balancing of Environmental Systems.

1.3 Definitions

- A. Adjusting: Varying of system flow by modifying settings of dampers and valves, in combination with varying fan speeds to obtain optimum operating conditions for the entire system.
- B. Balancing: Proportioning of air and hydronic flows through system mains, branches and terminal devices using standardized procedures to obtain specified air or hydronic flow while imposing the least amount of restriction on the HVAC system.
- C. Testing: Use of specialized and calibrated instruments to measure temperatures, pressures, rotational speeds, electrical characteristic, air and hydronic flow in velocities or quantities used in evaluating the performance of an HVAC system.

1.4 Coordination

- A. The testing, adjusting and balancing Contractor shall coordinate his work with the mechanical system and temperature control system installing Contractors to accomplish coordination and verification of system operation and readiness for testing, adjusting and balancing.

- B. Coordinate and assist CxP with all verification activities including providing all required sampling data necessary for the commissioning process.

1.5 Submittals

A. Qualification Statements:

1. Submit company's certification documents, including:
2. Contractor Certification:
3. Supervisor Certification
4. Technician Certification
5. Submit name of testing agency to Owner within thirty (30) days on Notice to Proceed.
6. Submit list of projects completed by testing agency of similar size, scope and equipment. Include name of Contractor and building Owner contacts.
7. Submit a certification letter stating that the TAB agency is an independent entity not owned in part or in whole by any subcontractor employed on the current project.

B. Reports:

1. Deficiency Report: Following examination of installed system, prior to balancing, submit report indicating system deficiencies that would prevent proper testing, adjusting and balancing of systems and equipment to meet specified performance.
2. TAB Report: Submit a copy of the complete testing, adjusting and balancing report to FMC Project Manager and RECS Atlanta Staff Engineer via email when it becomes available. Report shall include any drawings indicating air outlets, thermostats and equipment identified to correspond with data sheets.
3. Reports shall be on TABB/SMACNA (NEBB or AABC), forms that indicate information addressing each of the testing methods, readings and adjustments.

C. Closeout Submittals:

1. Provide complete copy of testing, adjusting and balancing report. Include report in operation and maintenance manual.

1.6 Quality Assurance

A. Qualifications:

1. Testing and balancing shall be performed by a testing agency who specializes in testing, adjusting and balancing of heating, ventilating, air-moving equipment, air-conditioning systems and hydronic systems, and has a minimum of one (1) year experience.
2. Testing agency shall have successfully completed a minimum of five (5) projects, similar in size and scope.

3. Testing agency shall be a certified member of TABB (AABC and/or NEBB).
4. Maintain a copy of applicable standards at the project site.

B. Certifications:

1. TAB Technician shall be certified by a nationally recognized certifying agency (AABC and/or NEBB).
2. Perform total system balance in accordance with Testing, Adjusting and Balancing Bureau (TABB) – Quality Assurance Program for Environmental Systems Balance, and (AABC National Standards for Field Measurement and Instrumentation and/or NEBB Quality Assurance Program – Conformance Certification).

C. Project Conditions

1. Testing, adjusting and balancing shall commence after the HVAC systems installation is complete and in working order. Associated areas of general construction shall be in place including interior and exterior doors, windows, walls, ceilings and existing conditions.

D. Special Warranty

1. Provide warranty for period of ninety (90) days following physical occupancy of building, during which time the Owner may request a re-check of up to 10% of total number of terminals, or resetting of any outlet, coil or device listed in the test report. This period of time shall be no longer than 180 days after submission of the completed report.
2. Warranty shall meet the requirements of the following program(s):
 - a. TABB – Quality Assurance Program
 - b. AABC – National Performance Guarantee
 - c. NEBB – Conformance Certification

PART 2 - PRODUCT

PART 3 - EXECUTION

- 3.1 Prior to commencing testing, adjusting and balancing of environmental system(s), verify the following conditions; if deficiencies are evident, submit Deficiency Report to Engineer. Do not begin testing, adjusting and balancing of environmental system until deficiencies have been remedied.
 - A. Systems are started and operating in a safe and normal condition.
 - B. Temperature control systems are installed, complete, and operable.
 - C. Automatic and manual dampers are operable and fully open.
 - D. Thermal overload protection is in place for fans, pumps, chillers and other equipment.

- E. Start up air filters are removed.
- F. Final filters are clean and properly installed.
- G. Duct and fan systems are clean.
- H. Fans are rotating correctly.
- I. Fire and volume dampers are in place and open.
- J. Air coils fins are cleaned and combed.
- K. Access doors are closed and duct end caps are in place.
- L. Air outlets are installed and connected.
- M. Hydronic systems are pressure tested, flushed, filled and properly vented.
- N. Leak testing on duct system has been performed in accordance with SMACNA Standards, or as specified.
- O. Pumps are rotating correctly.
- P. (Start-up/construction) strainers have been removed and all permanent strainers are clean and in place.
- Q. Gauges and/or test parts are properly located for balancing.
- R. Service and balance valves are fully open.

3.2 Site Tolerances

- A. Air Handling Systems: Adjust to within plus 10 percent of outlet total plus allowable leakage rate.
- B. Air Outlets and Inlets: Adjust to within plus or minus 10 percent of design for the space.
- C. Hydronic Systems: Adjust to within plus or minus 10 percent of design flow.
- D. Hydronic Terminal Devices: Adjust to within plus or minus 10 percent of design flow.

3.3 Air System Procedure

- A. Adhere to the following procedure:
 - 1. TABB – HVAC Testing, Adjusting and Balancing International Standards; with particular focus on the following chapters:
 - 2. Preliminary TABB procedures
 - 3. General air systems TABB procedures

- B. TABB procedures for specific (VAV, CAV, Multizone, Dual duct, etc.) air systems
- C. Sheet Metal and Air Conditioning Contractors' National Association (SMACNA) HVAC Systems – Testing, Adjusting and Balancing.
- D. NEBB – Procedural standards for TAB of environmental systems.
- E. AABC – National standards for total systems balance.
- F. Minimum air procedures should include the following:
 - 1. Test and adjust fan RPM to design requirements.
 - 2. Test and record motor full load nameplate rating and actual ampere draw.
 - 3. Test and record system static pressures, fan suction and discharge.
 - 4. Adjust all main supply and return air duct to within tolerances listed in this section of work
 - 5. Test and adjust each diffuser, grille and register. Reading and tests of diffusers, grilles and registers shall include design velocity (FPM) and adjusted velocity, design CFM and adjusted CFM.
 - 6. Test and record outside, mixed air, and discharge temperatures (D.B. for heating cycle, D.B. and W.B. for cooling cycle).
 - 7. In coordination with the ATC contractor, set adjustments of automatically operated dampers to operate as specified, indicated and/or noted.
 - 8. Test and adjust air handling and distribution systems to provide required or design supply, return, outside and exhaust air quantities within design tolerance
 - 9. In air systems employing filters, blank off filter area to simulate a pressure drop that is midway between that of a clean filter and that of a dirty filter.
 - 10. Make air velocity measurements in ducts by Pitot tube traverse entire cross-sectional area of duct in accordance with SMACNA equal area method or Log Linear method.
 - 11. Measure air quantities at all air inlets and outlets.
 - 12. Use volume control devices to regulate air quantities only to the extent that adjustments do not create objectionable air motion or sound levels.
 - 13. Vary total system air quantities by adjustments of fan speeds. Provide drive changes recommendations. Vary branch air quantities by damper regulation.
 - 14. Measure static air pressure conditions on air supply units, including filter and coil pressure drops, and total pressure across the fan. Make allowances for loading of filters and coils.

15. Adjust outside air automatic dampers. Outside air, return air and exhaust dampers for design conditions within specified tolerances.
16. Where modulating dampers or economizers are provided, take and record measurement at full return air, minimum outside air and 100 percent outside are mode of operation.
17. Verify and record, in the T&B Report, “K” factors for all VAV air terminal devices and air flow stations.

3.4 Hydronic System Pressure

A. Adhere to the following procedure:

1. Testing, Adjusting and Balancing Bureau (TABB) – International Standards for Environmental Systems Balance
2. SMACNA – HVAC Testing, Adjusting and Balancing International Standards; with particular focus on the following chapter:
 - a. Hydronic TAB procedures
 - b. NEBB – Procedural standards for TAB of environmental systems.
 - c. AABC – National standards for total systems balance.

B. Hydronic balancing shall include the following minimum data:

1. Prepare itemized equipment schedules, listing all heating and/or cooling elements and equipment in the systems to be balanced. List, in order on equipment schedules, by pump or zone according to the design, all heating and/or cooling elements, all zone balancing valves, and circuit pumps, ending with the last items of equipment or transfer element in the respective zone or circuit. Include on schedule sheet column titles listing the location, type of element or apparatus, design conditions and measured conditions. Prepare individual pump report sheets for each zone or circuit.
2. Use calibrated Venturi tubes, orifices, metered fittings, pressure gages and direct reading instrumentation to determine flow rates for system balance. Where flow-metering devices are not installed, flow balance in temperature difference across various heat transfer elements in the system is acceptable.
3. Adjust systems to provide specified pressure drops and flows through heat transfer elements prior to thermal testing. Perform balancing by measurement of temperature differential in conjunction with air balancing.
4. Adjust hydronic distribution systems by means of balancing cocks, valves and fittings. Do not use service or shut-off valves for balancing unless indexed for balance point.
5. Test pumps and adjust flow. Record the following on pump report sheets:
 - a. Suction and discharge pressure;

- b. Running amps and brake horsepower of pump motor under full flow and no flow conditions;
 - c. Pressure drop across pump in feet of water and total GMP pump is handling under full flow conditions.
6. Where available pump capacity is less than total flow requirements or individual system parts, proportional balancing must be performed.

3.5 Adjusting

- A. Recorded data shall represent actual measured or observed conditions.
- B. Permanently mark setting of valves, dampers and other adjustment devices allowing for settings to be restored. Set and lock memory stops.
- C. Leave systems in proper working, replacing belt guards, closing access doors, closing doors to electrical switch boxes and restoring thermostats to specified settings.
- D. Areas or rooms designed to maintain positive, negative or balanced air pressures with respect to adjacent spaces, as indicated by the design air quantities, require special attention. Adjust fan drives, distribution dampers, terminals and controls to maintain indicated pressure relationship.

END OF SECTION 230593

SECTION 230700 - INSULATION

PART 1 - GENERAL

- 1.1 This section includes insulation for piping, ductwork, and equipment, as shown on the plans.
- 1.2 All insulation, linings, coverings, and adhesives shall have a flame spread classification of 25 or less and a smoke developed rating of not more than 50.
- 1.3 Insulation shall be Certainteed, Owens Corning, or Johns-Manville.

PART 2 - PRODUCT

2.1 Duct

- A. Unless otherwise noted in the drawings all rectangular and round air conditioning supply, return, exhaust, and outside air duct shall be externally insulated with 3" thick, 3/4 lb. density foil scrim Kraft jacketed insulation. Joints shall be wrapped with a minimum of 3" wide FSK band of insulation to prevent any possible leakage and condensation. Ducts with widths over 30" shall be further secured on the underside with mechanical fasteners on 18" maximum centers.
- B. In addition to the duct wrap specified in B1.a of this specification, all low pressure rectangular supply and return ductwork shall be lined for 15 feet downstream from air handling unit (or up to and including the first 90 degree elbow). Duct liner shall be 1" thick, 2lb. dense, Shuller Permorate Linacoustic HP, or approved equivalent. Coat all exposed leading edges and transverse joints with a fire retardant adhesive.
- C. Duct sizes shown are actual duct dimension. Where ductwork is lined, as noted above, the duct insulation thickness shall be added to the listed ductwork dimensions for final duct size.
- D. Ductwork located in mechanical rooms shall be wrapped with duct board insulation 2" thickness rigid Fiberglas Owens/Corning or equal, ASTM C 612, 3 pounds per cubic foot density, with Foil reinforced jacket. The board shall be attached with field applied perforated base pins or weld pins applied on 12" centers. Finish shall be 8oz canvas jacket, totally sized with Foster 81-42W or equal lagging adhesive. Corner board shall be used on all edges.

2.2 Piping

- A. All heating hot water supply and return piping, fittings, valves, elbows etc., above grade shall be insulated with 4 lb. density snap-on type glass fiber pipe insulation in molded sections with factory applied all service jacket. Seams shall be closely buttoned together and secured by self-sealing or pasting the all service lap. Fittings insulation shall be milled pre-fabricated of same material and thickness as on adjacent pipe. Exposed fittings shall be finished with 8 oz. canvas jacket neatly pasted in place or factory pre-formed PVC jacket covers. Piping 2" or less shall have 1" thick insulation. Piping larger than 2" shall have 2" thick insulation. Piping exposed in occupied spaces and mechanical equipment rooms shall have canvas lagged in place for painting.
- B. All condensate drain piping, make-up water piping, all refrigerant suction piping, and all refrigerant piping exposed on the exterior of the building shall be insulated with 1.5" wall

tubular closed cell elastomeric insulation with all joints butted and cemented tight. Insulation shall be Rubatex R-180-FS or equal. Cover exterior insulation with aluminum jacket.

2.3 Equipment

- A. All equipment associated with the chilled water operation such as air separator, chiller (if not factory insulated), etc. shall be insulated with one inch thick 6 p.c.f. dense foam rubber insulation such as Armstrong Armaflex 22 or approved equal. The insulated surfaces shall be covered with eight-ounce canvas pasted in place and glue sized twice for painting. Canvas shall be coated twice with Foster fireproof lagging to assure specified flame and smoke spread ratings.
- B. All equipment associated with the hot water operation such as air separators, converters, etc. shall be insulated with 2 ½" thick, high density fiberglass insulation such as Ultralite or approved equivalent by Owens-Corning or Johns-Manville. The insulated surface shall be covered with eight-ounce canvas as described in Paragraph B.1 about.
- C. All equipment associated with steam such as steam to hot water converter, etc., shall be insulated with 2" of calcium silicate insulation and covered with 30 lb. felt covering.

PART 3 - EXECUTION

- 3.1 Insulation shall be installed in accordance with manufacturer's recommendations.
- 3.2 All exterior piping insulation above grade shall be provided with a protective aluminum jacket with a factory-applied asphalt and kraft paper moisture barrier. Aluminum jackets shall be cross-crimped (longitudinally corrugated) for strength. Aluminum jackets shall be not less than 0.106" thick and shall be secured with aluminum or stainless steel screw; not more that 8" apart.
- 3.3 All piping exposed outdoors shall be wrapped with electric trace before insulation is applied. Heat trace shall be equal be self-regulating with hard-wired power connection. See plans for wattage and voltage.
- 3.4 No chilled water shall be allowed to circulate prior to completion of insulation.
- 3.5 Any pipe covered prior to leak testing shall be exposed at contractor expense.
- 3.6 All piping shall be provided with identification in accordance with ANSI A13.1-1981 standards. Markers shall be located at each wall, floor or ceiling penetration, and at every 20 ft. Markets shall be fully legible from floor level showing medium contained pipe, and direction flow. Wording on markers shall be as follows:
 - A. Hot Water Supply
 - B. Hot Water Return
- 3.7 Provide 18 gauge 12" length sheet metal saddle at all hangers.
- 3.8 No insulation thickness shall be reduced at hanger.

END OF SECTION 230700

SECTION 230900 – INSTRUMENTATION AND CONTROL FOR HVAC

Project: Green Road HVAC Upgrade

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SECTION 230900 – INSTRUMENTATION AND CONTROL FOR HVAC

PART 1 – GENERAL

Contractor Responsibility

The FMCS Contractor shall be fully responsible for reading, understanding, and executing all requirements contained in the **Engineering Services – Instrumentation and Control Specification for HVAC**, Project: **Green Road HVAC Upgrade**. The Contractor must follow all requirements, procedures, coordination steps, integration standards, and device compatibility rules specified throughout **Section 230900**, including all supplemental Green Road project requirements in **Section 1.31**.

A. Required BAS Platform – Vykon JACE-9000 Only (No Substitutes Allowed)

The Contractor shall ensure that **only Vykon JACE-9000 Network Area Controllers (NACs)** are installed on the project. **No substitutions** are permitted under any circumstances. This requirement exists because the City of Raleigh BAS is a Niagara enterprise implementation leveraging the Vykon open architecture, which includes:

1. Open API and Open Protocol support
2. Open distribution model
3. Ability to integrate third-party drivers and modules
4. No proprietary licensing bundles

5. No proprietary or locked manufacturer ecosystems
6. No forced use of vendor-specific programming tools

This ensures long-term interoperability, flexibility, and enterprise scalability.

B. Supported IP Controllers (Strict List – No Substitutes Without Approval)

The specification strictly limits acceptable IP controllers to the following:

1. **Vykon Edge**
2. **Honeywell Optimizer**
3. **Lynxspring JENEsys Edge Controller**

For VAV units, acceptable IP controllers are:

1. **Vykon Edge VAV**
2. **Honeywell Optimizer VAV**
3. **Lynxspring JENEsys Edge VAV IP Controller**

Only these controller families are permitted unless the **Engineering Services Technology Manager** explicitly approves a substitution.

C. Communication Requirements – BACnet/IP Only

The FMCS Contractor shall implement **BACnet/IP as the sole communication protocol** for all HVAC controls. **BACnet MSTP is not allowed** except when integrating a factory-supplied mechanical equipment controller that **only** supports MSTP. Under all other conditions, MSTP is prohibited.

Additionally:

1. **Power and gas meters shall use BACnet/IP**
2. **Modbus is not permitted** for metering devices

D. Required Site Survey – Bid Phase

During the bidding phase, the Contractor shall perform a **site survey** with coordination from the City's Building Automation Specialist to:

1. Determine available City network access
2. Identify the number of required switch ports for the project
3. Validate infrastructure support for all BACnet/IP controllers

E. Section 1.31 – Green Road Supplemental Specifications (Mandatory)

The Contractor shall follow all Green Road-specific requirements listed in **Section 1.31**, including:

1. Replacement of JACE-8000 with JACE-9000
2. Implementation of Niagara 4.15
3. BACnet/IP enforcement
4. Secure FOXs client connections
5. Proper alarm class configurations
6. Graphics review procedures
7. Point list, tagging, trend setup
8. Freeze protection safety programming
9. Supervisor-level history uploads
10. Dashboard configuration
11. Provisioning backups

F. Section 1.6 – Submittal Requirements (Contractor Must Deliver Accurately)

The Contractor must pay **close attention to Section 1.6 – Submittals**, which outlines detailed requirements for all documentation, drawings, device cut sheets, architecture diagrams, point lists, sequences of operation, trunk cable schematics, and GUI graphics.

The Contractor shall:

1. Deliver **accurate and complete** project submittals as specified
2. Provide **eight copies** of all required shop drawings
3. Include **manufacturer cut sheets**, wiring diagrams, BACnet network architecture, device lists, point tables, alarm lists, and system-level documentation
4. Provide **AutoCAD/Visio**-ready as-built drawings at project completion
5. Ensure all submittals reflect correct model numbers, configurations, protocols, addressing, and integration paths
6. Include **graphics (80%–90% complete)** with navigation flowcharts as required

Accurate submittal delivery is essential to City acceptance, and any incomplete, missing, or incorrect documentation may result in rejection, rework, and delays in project approval.

1.1 SUMMARY

- A. This section describes the scope of work for the Facility Management and Control System that must be installed by a qualified FCMS Contractor and integrated to the Enterprise Server by the Enterprise Developer. This section also coordinates the responsibilities of the Mechanical and Electrical trade contractors pertaining to control products or systems, furnished by each trade, and that will be integrated by this Division.
- B. All labor, material, equipment and software not specifically referred to herein or on the plans, that is required to meet the functional intent of this specification, shall be provided without additional cost to the Owner.
- C. It is the owner's goal to implement an open system that will allow products from various suppliers to be integrated into a unified system. Allowing various system integrators the ability to maintain, service and perform work on NAC and all controllers that are integrated into the system in order to provide flexibility and expansion of the system. The Owner shall be the named license holder of all software associated with any and all incremental work on the project(s).

1.2 SCOPE OF WORK

- A. The Facility Management and Control System (FMCS) shall be comprised of Network Area Controllers (NAC) within each facility. The NAC shall connect to the owner's local or wide area network, depending on configuration. Access to the system by and End User shall be accomplished, either locally in each building, or remotely from a central site or sites, shall be accomplished through a standard Web browser via the Internet and/or local area network. Authorized users shall also be able to configure a dashboard view of pertinent data and this view shall be saved for later use. Each NAC shall communicate to unitary BACnet IP controllers (BIPC), Niagara IP Controllers (NIPC) or MODBUS controllers on equipment such as air handling units, VAV boxes, switchgear, etc. or provided by the FCMS Contractor. The NAC shall also connect to other open and legacy protocol systems devices provided under other

Divisions. It is the owner's goal to eliminate any gateway or redundant device(s) between NAC and IP controllers.

- B. The Facility Management and Control System (FMCS) as provided in this Division shall be the Vykon JACE (NAC) based on the Niagara4 Framework (or "Niagara4"), a device framework developed by Tridium. Vykon provides an open automation infrastructure, an open license, and is available from multiple systems integrators. Vykon integrates diverse systems and devices (regardless of manufacturer, communication standard or software) into a unified platform that can be easily managed in real time over a secure network using a standard Web browser.
- C. The work provided in this specification shall be performed by two entities. The FMCS Contractor shall have overall responsibility for the Division work. The Enterprise Developer shall be appointed by the Owner and shall provide all work at the Enterprise Server level. The successful general contractor shall include all costs of the Enterprise developer in their bid documents. Owner will oversee all work of the Enterprise Developer and services they provide. See Section 1.3 for more detail on the division of work.
- D. All materials and equipment used shall be standard components. All systems and components shall have been thoroughly tested and proven in actual use for at least two years.
- E. All wiring shall be done in accordance with all local and national codes.

1.3 DIVISION OF WORK

- A. The Division 23 FMCS contractor shall be responsible for all Network Area Controllers (NAC), Local IP controllers (LIPC), any miscellaneous third-party factory mounted equipment controllers, control devices, control panels, controller programming, controller programming software, controller input/output and power wiring and controller network wiring specified to be provided in Division 23.
- B. The Division 26 (if applicable) contractors shall be responsible for all Security NAC, miscellaneous control devices, control panels, controller programming, controller programming software, controller input/output and power wiring and controller network wiring specified to be provided in Division 26. These devices shall be configured and commissioned by Division 26 contractors and later managed in the NAC by FMCS contractor. Exception - Security NACs provided in Division 26 shall be connected to the Enterprise Server by the Enterprise Developer.
- C. The FMCS contractor shall also be responsible for the software and programming of the NAC, graphical user interface software (GUI), User Configurable Dashboard software, and connection of the NAC to the local or wide area network. FMCS shall be responsible for development of all graphical screens, Web browser pages, setup of schedules, logs and alarms, and network management for all LIPCs, and other third party devices provided in Division 23 and 26. Any third party devices not provided by FMCS contractor shall be configured and commissioned by appropriate contractor and later managed in the NAC or LIPC by FMCS contractor.
- D. For reasons of security and consistency, it is the owner's intention to divide the work defined in this section into two sections. Work performed at the NAC level and below shall be performed by a pre-qualified FMCS Systems Integrator. All work provided at the Enterprise Server and between the server and other systems shall be provided by the owner appointed Enterprise Developer. The Enterprise Developer shall be responsible for the "learning" of the WBI (web

browser graphics) from the NAC to the Enterprise Server, the configuration of the Periscope Dashboard software and the global integration strategies across NACs and other intelligent building systems. The Enterprise Developer shall also be responsible for all Security integration at the Server level, if applicable.

- E. The Enterprise Developer shall be responsible for all Server upgrades and ongoing maintenance licensing for Server and NACs.

1.4 RELATED WORK SPECIFIED ELSEWHERE

- D. A. Products integrated and installed but not furnished under this section

- 1. Project specific equipment

- a. NAC shall be a JACE-9000 w/ minimum 2GB LPDDR4 RAM, 2 isolated RS-485 ports, 2 10/100/1000 MB Ethernet ports, Tridium's standard drivers, USB Backup and Restore and Wi-Fi connectivity.
- b. Removeable micro-SD card with 8GB total storage
- c. USB type C connector debug port
- d. Power Supply Requirements
 - 24VAC rated at 24VA minimum or 24VDC rated at 1A (24W) minimum.
- e. WPM-8000 universal power supply for NAC
- f. Provide additional non-standard communications drivers per specifications.
- g. Provide additional RS-485, RS-232 and IO modules as needed to connect to BAS and third-party systems or networks.
- h. Add Device pack license as needed to communicate with FMCS devices. NAC shall have the capacity to connect to a minimum of 25 devices and should be sized to provide a minimum of 20% spare capacity
- i. Provide initial 18-month Software Maintenance Agreement on all NACs.

1.5 QUALITY ASSURANCE

- A. The FMCS system shall be designed and installed, commissioned and serviced by factory trained personnel. Systems Integrator shall have an in-place support facility within 100 miles of the site with technical staff, spare parts inventory and necessary test and diagnostic equipment.
 - 1. The Systems Integrator shall provide full time, on site, experienced project manager for this work, responsible for direct supervision of the design, installation, start up and commissioning of the FMCS.
 - 2. The Bidder shall be regularly engaged in the manufacturing, installation and maintenance of FMCS systems and shall have a minimum of ten (10) years of demonstrated technical expertise and experience in the manufacture, installation and maintenance of FMCS systems similar in size and complexity to this project with a maintained service organization. Provide a list of 10 projects, similar in size and scope to this project, completed within the last five years.
- B. Materials and equipment shall be the catalogued products of manufacturers regularly engaged in production and installation of automatic temperature control systems and shall be manufacturer's latest standard design that complies with the specification requirements.

- C. All FMCS peer-to-peer network controllers, central system controllers and local user displays shall be UL Listed under Standard UL 916, category PAZX; Standard ULC C100, category UUKL7; and under Standard UL 864, categories UUKL, UDTZ, and QVAX and be so listed at the time of bid. All floor level controllers shall comply, at a minimum, with UL Standard UL 916 category PAZX; Standard UL 864, categories UDTZ, and QVAX and be so listed at the time of Bid.
- D. All electronic equipment shall conform to the requirements of FCC Regulation, Part 15, Governing Radio Frequency Electromagnetic Interference and be so labeled.
- E. The manufacturer of the building automation system shall provide documentation supporting compliance with ISO-9002 (Model for Quality Assurance in Production, Installation, and Servicing) and ISO-14001 (The application of well-accepted business management principles to the environment). The intent of this specification requirement is to ensure that the products from the manufacturer are delivered through a Quality System and Framework that will assure consistency in the products delivered for this project.
- F. This system shall have a documented history of compatibility by design for a minimum of 15 years. Future compatibility shall be supported for no less than 10 years. Compatibility shall be defined as the ability to upgrade existing field panels to a compatible level of technology, and extend new field panels on a previously installed network.
 - 1. Compatibility shall be defined as the ability for any existing field panel microprocessor to be connected and directly communicate with new field panels without bridges, routers or protocol converters.

1.6 SUBMITTALS

- A. Provide individuals experienced with the installation and startup of equipment related to this type of integration.
 - 1. Eight copies of shop drawings of the entire FMCS shall be submitted and shall consist of a complete list of equipment and materials, including manufacturers catalog data sheets and installation instructions. Shop drawings shall also contain complete wiring and schematic diagrams, software descriptions, calculations, and any other details required to demonstrate that the system has been coordinated and will properly function as a system. Terminal identification for all control wiring shall be shown on the shop drawings. The FMCS Systems Integrator shall submit an architecture layout that depicts devices from the Server to NAC down to the device level.
 - 2. A complete written Sequence of Operation shall also be included with the submittal package. The FMCS Systems Integrator shall coordinate data from other contractors supplying products and systems, as part of their package and shall provide catalog data sheets, wiring diagrams and point lists to the owner for proper coordination of work.
- B. Submittal shall also include a trunk cable schematic diagram depicting operator workstations, control panel locations and a description of the communication type, media and protocol. The FMCS Systems Integrator shall be responsible for integrating all network level devices into the overall trunk cable schematic diagrams for the entire Wide Area Network (WAN).

- C. Submittal shall also include a complete point list of all points to be connected FMCS by the Systems Integrator. FMCS System Integrator shall provide necessary point lists, protocol documentation, and factory support information for systems provided in their respective divisions but integrated into the FMCS.
 - D. The submittal shall include a final copy of the sequence of operations.
 - E. Submittal shall also include a copy of each of the graphics developed for the Graphic User Interface including a flowchart (site map) indicating how the graphics are to be linked to one another for system navigation. The graphics are intended to be 80% - 90% complete at this stage with the only remaining changes to be based on review comments from the A/E design team and/or Owner. Submittal shall also include a copy of the expected Dashboard viewlets being provided for owner configuration. It is expected that the successful Systems Integrator shall utilize the City of Raleigh graphic templates as much as possible. The owner will provide an example of an acceptable graphic template. Where a particular graphic template does not exist, the Integrator shall create a similar template and gain approval during submittal process.
 - F. Upon completion of the work, provide a complete set of 'as-built' drawings and application software on compact disk. Drawings shall be provided as AutoCAD™ or Visio™ compatible files. Eight copies of the 'as-built' drawings shall be provided in addition to the documents on compact disk. Division 23 and 26 contractors shall provide as-builts for their portions of work. The FMCS Systems Integrator shall be responsible for as-builts pertaining to overall FMCS architecture and network diagrams. All as built drawings shall also be installed into the FMCS server in a dedicated directory.
 - G. The submittal shall include manufacturer cut sheets for every physical device furnished, including controllers, sensors, actuators, valves, dampers, meters, control panels, and other field devices. Submitted literature shall clearly indicate the specific model and configuration being provided.
 - H. The submittal shall include an integration plan outlining BACnet network architecture, including device addresses, network communication requirements, and integration procedures for all third-party systems. The plan shall identify responsible parties, required interfaces, and coordination requirements necessary to ensure successful system integration.
 - I. The submittal shall provide documentation confirming that all controller licenses, including point capacity and software features, are properly sized to accommodate the complete BAS system point count. The submitted licensing shall match or exceed the required number of points for each controller.
 - J. The submittal shall provide site operating schedules for all BAS-controlled systems, including equipment start/stop times, occupancy schedules, holiday schedules, and any seasonal or special operating requirements. Schedules shall be coordinated with the Owner's operational requirements prior to system programming.
- 1.7 NIAGARA STATION BACKUP
- A. At the completion of the project, the contractor shall provide the City of Raleigh with a backup of the Niagara Station

1.8 SPECIFIC NOMENCLATURE

A. Acronyms used in this specification are as follows:

FMCS	Facility Management and Control System
TCS	Temperature Control System
NAC	Network Area Controller (Vykon JACE 9000)
LIPC	Local IP Controllers (includes BIPC, NIPC and MIPC)
BIPC	BACnet IP Controller
NIPC	Niagara IP Controller
MIPC	Modbus IP Controller
GUI	Graphical User Interface
WBI	Web Browser Interface
POT	Portable Operator's Terminal
PMI	Power Measurement Interface
DDC	Direct Digital Controls
LAN	Local Area Network
WAN	Wide Area Network
OOT	Object Oriented Technology
PICS	Product Interoperability Compliance Statement

1.9 SOFTWARE LICENCE AGREEMENT

- A. The Owner shall sign a copy of the manufacturer's standard software and firmware licensing agreement as a condition of this contract. Such license shall grant use of all programs and application software to Owner as defined by the manufacturer's license agreement but shall protect manufacturer's rights to disclosure of trade secrets contained within such software.
- B. It is the owner's express goal to implement an open system that will allow products from various suppliers to be integrated into a unified Vykon system in order to provide flexibility for expansion, maintenance, and service of the system. The Owner shall be the named license holder of all software associated with any and all incremental work on the project(s). In addition, the Owner shall receive ownership of all job specific configuration documentation, data files, and application-level software developed for the project. This shall include all custom, job specific software code and documentation for all configuration and programming that is generated for a given project and/or configured for use with the NAC, FMCS Server(s), and any related LAN / WAN / Intranet and Internet connected routers and devices. Any and all required IDs and passwords for access to any component or software program shall be provided to the owner.

1.10 DELIVERY, STORAGE AND HANDLING

- A. Provide factory-shipping cartons for each piece of equipment and control device. Maintain cartons through shipping, storage, and handling as required to prevent equipment damage. Store equipment and materials inside and protected from weather.

1.11 JOB CONDITIONS

- A. Cooperation with Other Trades: Coordinate the Work of this section with that of other sections to ensure that the Work will be carried out in an orderly fashion. It shall be this Systems Integrator's responsibility to check the Contract Documents for possible conflicts between his Work and that of other crafts in equipment location, pipe, duct and conduit runs, electrical outlets and fixtures, air diffusers, and structural and architectural features.

1.12 WARRANTY

- A. Provide all services, materials and equipment necessary for the successful operation of the entire FMCS for a period of one year after beneficial use.
- B. The adjustment, required testing, and repair of the system includes all computer equipment, transmission equipment and all sensors and control devices.
- C. With owner pre-approval, the on-line support services shall allow the local FMCS Systems Integrator to connect over City of Raleigh Virtual Private Network (VPN) to monitor and control the facility's building automation system. Pending owner approval, this remote connection to the facility shall be within 2 hours of the time that the problem is reported. This coverage shall be extended to include normal business hours, after business hours, weekends and holidays.
 - 1. If the problem cannot be resolved on-line by the local office, the national office of the building automation system manufacturer shall have the same capabilities for remote connection to the facility.
 - 2. If the problem cannot be resolved with on-line support services, the FMCS manufacturer shall dispatch the appropriate personnel to the job site to resolve the problem within a reasonable time frame.

1.13 ACCEPTABLE SYSTEM INTEGRATORS

- A. The FMCS Systems Integrator shall provide NAC hardware, software, local IP controllers (LIPC) and DDC components. NAC hardware and software shall be the Vykon brand.
- B. Local IP controllers shall be either Vykon (NIPC), Honeywell Optimizer Controller, Lynxspring JENEsys Edge Controller, or an owner-approved controller.
- C. The FMCS Systems Integrator shall have a technical support group accessible that is staffed with qualified personnel, capable of providing instruction and technical support service for networked control systems.
- D. The successful FMCS Systems Integrator shall not have password access to the Enterprise Server and shall be restricted to NAC access only.
- E. Acceptable Systems Integrators of the hardware and software components as specified herein are as follows:
 - 1. Mechanical Contractors
 - 2. Vykon by Energy Automation Technologies
 - 3. Honeywell by EMCOR Group

4. Honeywell by Bosch Building Technologies
5. Other – City Approved

PART 2 - PRODUCTS

1.14 GENERAL

- A. The Facility Management Control System (FMCS) shall be comprised of a network of interoperable, stand-alone digital controllers, a computer system, graphical user interface software, printers, network devices and other devices as specified herein.
- B. The installed system shall provide secure password access to all features, functions and data contained in the overall FMCS.

1.15 OPEN, INTEROPERABLE, INTEGRATED ARCHITECTURES

- A. The intent of this specification is to provide a peer-to-peer networked, stand-alone, distributed control system with the capability to integrate ANSI/ASHRAE Standard 135-2001 BACnet, MODBUS, , and other existing open and proprietary communication protocols if applicable in one open, interoperable system.
- B. The supplied computer software shall employ component-based technology (OOT) for representation of all data and control devices within the system. In addition, adherence to industry standards including ANSI / ASHRAE™ Standard 135-2001, BACnet to assure interoperability between all system components is required. For each BACnet device, the device supplier must provide a PICS document showing the installed device's compliance level. Minimum compliance is Level 3; with the ability to support data read and write functionality. Physical connection of BACnet devices shall be via Ethernet (BACnet IP). BACnet RS-485 (BACnet MSTP) solutions are not acceptable unless written approval by Owner has been received prior to bid date.
- C. All components and controllers supplied under this Division shall be true "peer-to-peer" communicating devices. Components or controllers requiring "polling" by a host to pass data shall not be acceptable.
- D. The supplied system must incorporate the ability to access all data using standard Web browsers without requiring proprietary operator interface and configuration programs. An Open Database Connectivity (ODBC) Systems requiring proprietary database and user interface programs shall not be acceptable.
- E. An IP topology is required to manage DDC Controllers and the flow and sharing of data without unduly burdening the customer's internal Intranet network. FMCS Contractor shall provide/install a separate isolated IP network including cabling, switches, routers, power supplies, cabinets, etc. for the BIPC, NIPC and MIPC devices to communicate back to a local NAC. Only NACs will reside on the City's WAN/LAN.
 1. Maximum acceptable response time from any alarm occurrence (at the point of origin) to the point of annunciation shall not exceed 5 seconds for network connected user interfaces.

2. Maximum acceptable response time from any alarm occurrence (at the point of origin) to the point of annunciation shall not exceed 60 seconds for remote or dial-up connected user interfaces.

1.16 NETWORK

- A. The Local Area Network (LAN) shall be a minimum 100 Megabits/sec Ethernet network supporting BACnet, Java, XML, HTTP/HTTPS, and OBIX for maximum flexibility for integration of building data with enterprise information systems and providing support for multiple Network Area Controllers (NACs), user workstations and, if specified, a local server.
- B. Local area network minimum physical and media access requirements:
 1. Ethernet; IEEE standard 802.3
 2. Cable; 100 Base-T, UTP-8 wire, category 6
 3. Minimum throughput; 100 Mbps.

1.17 NETWORK ACCESS

- A. Remote Access.

For Local Area Network installations, the Owner shall provide a connection to the Internet to enable access via the City's Intranet to a corporate server. FMCS Systems Integrator shall connect to IP drop provided by the Owner within 25 feet.

1.18 NETWORK AREA CONTROLLER (NAC)

- a. The FMCS Systems Integrator shall supply one or more Network Area Controllers (NAC) as part of this contract to manage devices/points in all specification sections except for Division 280000 Security. Security NACs are provided under Division 280000 and all card access, video and intrusion detection shall be integrated into the existing Enterprise software by the Division 280000 Systems Integrator. This division shall be required to integrate BACnet zone information provided by the Division 280000 Systems Integrator into the HVAC and Lighting Sequence of Operation. The number of NACs provided by this Systems Integrator is dependent on the type/quantity of devices and points. It is the responsibility of the FMCS Systems Integrator to coordinate with all Division contractors to determine the quantity and type of NACs needed to fulfill the operating sequences.
- b. The Network Area Controller (NAC) shall provide the interface between the LAN or WAN and the field control devices and provide global supervisory control functions over the control devices connected to the NAC. It shall be capable of executing application control programs to provide:
 1. Calendar functions
 2. Scheduling
 - a. The contractor shall contact the owner to configure the Occupied and Unoccupied hours.

3. Trending
 - a. The contractor shall configure all trends according to the City of Raleigh's point naming and tagging requirements as outlined in the supplemental spreadsheet.
 4. Alarm monitoring and routing.
 5. Overrides
 - a. The contractor shall configure all overrides according to the specifications outlined here:
 - i. The Operator role shall not have 'Set' capability; only the Administrator role shall have 'Set' capability.
 - ii. The Operator role shall have 'Override' and 'Auto' capabilities.
 - iii. The 'Override Duration: Permanent' option shall be removed and replaced with a maximum override duration of 72 hours.
 6. Time synchronization
 - a. The contractor shall configure the Niagara Services > PlatformServices > NtpPlatformServiceQnx. The contractor shall contact the City of Raleigh Building Automation Specialist for the Time Server IP address.
 7. Haystack Tagging
 - a. The contractor shall configure haystack tagging according to the City of Raleigh's point naming and tagging requirements as outlined in the supplemental spreadsheet.
 8. Integration of BACnet IP/Modbus TCP/IP controller data
 9. Network Management functions for all third-party BACnet and Modbus based devices if specified as such.
 10. Provide license for a minimum of 25 devices connected.
- c. Point Naming and Haystack Tagging
- a. The contractor shall configure all point names, point tagging, and point trends according to the City of Raleigh's point naming and tagging requirements. This information shall be provided to the contractor in a supplemental spreadsheet.
 - b. The contractor shall provide a final list of points with names, Haystack tagging, and trends.
- d. The Network Area Controller must provide the following hardware features as a minimum (also see section 1.4.a):
1. Two Ethernet Port – 10/100/1000 Mbps
 2. One RS-232 port
 3. Two isolated RS-485 port
 4. USB Backup and Restore

5. Flash memory for long term data backup (If battery backup or flash memory is not supplied, the controller must contain a hard disk with at least 1 gigabyte storage capacity)
6. The NAC must be capable of operation over a temperature range of -20 to 60°C
7. The NAC must be capable of withstanding storage temperatures of between -40 to 85°C
8. The NAC must be capable of operation over a humidity range of 5 to 95% RH, non-condensing
- e. The NAC shall support standard Web browser access via the Intranet/Internet. It shall support a minimum of 32 simultaneous users.
- f. NAC Alarm Notification and actions
 1. The NAC shall provide alarm recognition, storage; routing, management, and analysis to supplement distributed capabilities of equipment or application specific controllers.
 2. The NAC shall be able to route any alarm condition to any defined user location whether connected to a local network or remote via dial-up telephone connection, or wide-area network.
 3. The contractor shall work with the City of Raleigh Building Automation Specialist to configure the correct alarm class for routing alarms to the supervisor station.
 - a. BACnet Alarms (Ping Failure)
 - i. The contractor shall configure the Ping Monitor as follows:
 1. Ping Enabled = true
 2. Ping Frequency = 05m
 3. Alarm On Failure = true
 4. Startup Alarm Delay = 10m
 - ii. The contractor shall use the alarm class "PingAlarm" for all BACnet alarms (Ping failures).
 - iii. The contractor shall route the PingAlarm class to the City_of_Raleigh Supervisor Station.
 4. Alarm generation shall be selectable for annunciation type and acknowledgement requirements including but limited to:
 - b. To alarm
 - c. Return to normal
 - d. To fault
 5. Provide timed (schedule) routing of alarms by class, object, group, or node.
 6. Provide alarm generation from binary object "runtime" and /or event counts for equipment maintenance. The user shall be able to reset runtime or event count values with appropriate password control.

7. Control equipment and network failures shall be treated as alarms and annunciated.
8. Alarms shall be annunciated in any of the following manners as defined by the user:
 - a. Screen message text
 - b. Email of the complete alarm message to multiple recipients. Provide the ability to route and email alarms based on:
 1. Day of week
 2. Time of day
 3. Recipient
 - c. Pagers via paging services that initiate a page on receipt of email message
 - d. Graphic with flashing alarm object(s)
9. The following shall be recorded by the NAC for each alarm (at a minimum):
 - a. Time and date
 - b. Location (building, floor, zone, office number, etc.)
 - c. Equipment (air handler #, accessway, etc.)
 - d. Acknowledge time, date, and user who issued acknowledgement.
 - e. Number of occurrences since last acknowledgement.
10. Alarm actions may be initiated by user defined programmable objects created for that purpose.
11. Defined users shall be given proper access to acknowledge any alarm, or specific types or classes of alarms defined by the user.
12. A log of all alarms shall be maintained by the NAC and/or a server (if configured in the system) and shall be available for review by the user.
13. Provide a “query” feature to allow review of specific alarms by user defined parameters.
14. A separate log for system alerts (controller failures, network failures, etc.) shall be provided and available for review by the user.
15. An Error Log to record invalid property changes or commands shall be provided and available for review by the user.
- g. NAC Data Collection and Storage
 1. The NAC shall have the ability to collect data for any property of any object and store this data for future use. See points list for required logs.
 2. The data collection shall be performed by log objects, resident in the NAC that shall have, at a minimum, the following configurable properties:
 - a. Designating the log as interval or deviation.

- b. For interval logs, the object shall be configured for time of day, day of week and the sample collection interval.
 - c. For deviation logs, the object shall be configured for the deviation of a variable to a fixed value. This value, when reached, will initiate logging of the object.
 - d. For all logs, provide the ability to set the maximum number of data stores for the log and to set whether the log will stop collecting when full, or rollover the data on a first-in, first-out basis.
 - e. Each log shall have the ability to have its data cleared on a time-based event or by a user-defined event or action.
- 3. All log data shall be archived to a database in the Enterprise Server and the data shall be accessed from a standard Web browser and the Periscope Dashboard.
- 4. All log data, when accessed from a server, shall be capable of being manipulated using standard SQL statements.
- 5. All log data shall be available to the user in the following data formats:
 - a. HTML
 - b. XML
 - c. Plain Text
 - d. Comma or tab separated values
- 6. Systems that do not provide log data in HTML and XML formats at a minimum shall not be acceptable.
- 7. The NAC shall have the ability to archive its log data remotely to a server on the network. Provide the ability to configure the following archiving properties, at a minimum:
 - a. Archive on time of day
 - b. Archive on user-defined number of data stores in the log (buffer size)
 - c. Archive when log has reached its user-defined capacity of data stores
 - d. Provide ability to clear logs once archive.
- h. NAC AUDIT LOG
 - 1. Provide and maintain an Audit Log that tracks all activities performed on the NAC. Provide the ability to specify a buffer size for the log and the ability to archive log based on time or when the log has reached its user-defined buffer size. Provide the ability to archive the log to a server. For each log entry, provide the following data:
 - a. Time and date
 - b. User ID
 - c. Change or activity: i.e., Change setpoint, add or delete objects, commands, etc.
- i. NAC DATABASE BACKUP AND STORAGE

1. The NAC shall have the ability to automatically backup its database. The database shall be backed up based on a user-defined time interval. Enterprise Developer shall coordinate with Owner to establish/implement a backup procedure.
2. Copies of the current database and, at the most recently saved database shall be stored in the NAC. The age of the most recently saved database is dependent on the user-defined database save interval.
3. The NAC database shall be stored, at a minimum, in XML format to allow for user viewing and editing, if desired. Other formats are acceptable as well, as long as XML format is supported.

1.19 INTEGRATED DEVELOPMENT ENVIRONMENT (IDE)

- A. An integrated development environment for development of graphic screens, control logic, security, alarm notification and data storage has been established using the Vykon Workbench Tool and currently resides on a Server in the existing datacenter and several laptops. The successful FMCS Systems Integrator shall utilize its own laptop for all programming and graphical development. The Enterprise Developer shall utilize the IDE at the server via a VPN connection or its own separate laptop IDE. The IDE residing on the central server shall be the most current version of the Vykon Workbench toolset and the FMCS Systems Integrator shall utilize the exact same version when programming NACs. Provide licenses for any new software configuration, programming or graphical tools that are required for viewing or managing the new BIPC, NIPC or MIPC devices.
- B. The server and NAC IDE tools shall be identical; however, it shall be possible to limit views and commands via a unique user profile and password in either. The IDE shall include a quick viewing of, and access to, the hierarchical structure of the database. Menu-pull downs, and toolbars shall employ buttons, commands and navigation to permit the operator to perform tasks with a minimum knowledge of the HVAC Control System and basic computing skills. These shall include, but are not limited to, forward/backward buttons, home button, and a context sensitive locator line (similar to a URL line), that displays the location and the selected object identification.
- C. Security. Each operator shall be required to log on to that system with a user name and password in order to view, edit, add, or delete data. The Owner shall control/set all passwords and security levels for all operators. The Owner shall provide the FMCS and Enterprise Developer with the standard passwords required to be used in the Enterprise Server and the NAC. The FMCS Systems Integrator shall not use any passwords except those provided by the owner. The system administrator shall have the ability to set passwords and security levels for all other operators. Each operator password shall be able to restrict the operators' access for viewing and/or changing each system application, full screen editor, and object. Each operator shall automatically be logged off of the system if no keyboard or mouse activity is detected.
- D. System Diagnostics. The system shall automatically monitor the operation of all workstations, modems, network connections, building management panels, and controllers. The failure of any device shall be annunciated to the operator.
- E. Alarm Management

1. The system will be provided with a dedicated alarm window or console. Refer to Sequence of Operations/Points List for Alarm strategies. The Alarm Console will notify the operator of an alarm condition and allow the operator to view details of the alarm and acknowledge the alarm. The use of the Alarm Console can be enabled or disabled by the system administrator. Alarms shall be created and grouped per the owner's requirements by the FMCS Systems Integrator at the NAC level. The Enterprise Developer shall bring the NAC alarms into the existing Enterprise server and generate the strategies to send alarms to the appropriate city or contractor parties.
2. Alarms shall be capable of being routed to any of the following:
 - a. Local Alarm Console (by FMCS Systems Integrator)
 - b. Remote Alarm Station (by Enterprise Developer)
 - c. Email recipient (multiple if needed) (by Enterprise Developer)
 - d. Local Printer connected to Personal Computer (by Enterprise Developer)
3. When the Alarm Console is enabled, a separate alarm notification window will supersede all other windows on the desktop and shall not be capable of being minimized or closed by the operator. This window will notify the operator of new alarms and un-acknowledged alarms. Alarm notification windows or banners that can be minimized or closed by the operator shall not be acceptable. Alarms shall be able to be mapped into groupings where the groupings have common displays, sounds or hyperlinks. This grouping shall be used to distinguish alarms when alarms are coming in from multiple sites or classes (i.e. buildings, regions, trades, etc.) for faster recognition.
4. The system shall be provided with an alarm database management view. The view shall allow a user with appropriate password to:
 - a. Filter or Clear old records before a certain date and time
 - b. Clear records older than the currently highlighted record
 - c. Clear all records
 - d. Modify the alarm table options including which alarm details are displayed, column width, etc.
 - e. Export the alarm database records to .pdf, text or CSV formats.

1.20 REAL-TIME WEATHER PROGRAM

- a. The contractor shall develop the logic to use the Outside Air Temperature and Outside Air Humidity sensors as the primary source of data provided to the BAS system or Niagara Station.
- b. The contractor shall develop the logic to use the Niagara Weather Service as the backup source for the Outside Air Temperature and Outside Air Humidity sensors.

2.7 WEB BROWSER CLIENTS

- A. The NAC system shall also allow use of an unlimited number of clients using a standard Web browser such as Internet Explorer™ or Chrome. The system shall be capable of providing a rich user experience (including full use of the engineering toolset) using java applets or a

simple user interface using only HTML5. Refer to Sequence of Operations for the client side display types that are required on this project.

- B. The Web browser software shall run on any operating system and system configuration that is supported by the Web browser. Systems that require specific machine requirements in terms of processor speed, memory, etc., in order to allow the Web browser to function with the FMCS, shall not be acceptable.
- C. The Web browser shall provide the same view of the graphics, schedules, calendars, logs, etc. as is provided by the Graphical User Interface. Systems that require different views or that require different means of interacting with objects such as schedules, or logs, shall not be permitted.
- D. The Web browser client shall support at a minimum, the following functions:
 - 1. User log-on identification and password shall be required. If an unauthorized user attempts access, a blank web page shall be displayed. Security using Java authentication and encryption techniques to prevent unauthorized access shall be implemented.
 - 2. Graphical screens developed for the GUI shall be the same screens used for the Web browser client (unless clearly stated in the sequence of operation). Any animated graphical objects supported by the GUI shall be supported by the Web browser interface. Enterprise Developer shall provide a FMCS Systems Integrator with a basis of performance/expectation for GUI. FMCS Systems Integrator shall use this standard graphic template or modify the graphics slightly to achieve the desired specification requirement/outcome.
 - 3. Storage of the graphical screens shall be in the Network Area Controller (NAC) and these graphics shall be “learned” by the Enterprise Server.
 - 4. Real-time values displayed on a Web page shall update automatically without requiring a manual “refresh” of the Web page.
 - 5. Owner shall have administrator-defined access privileges. Depending on the access privileges assigned, the user shall be able to perform the following:
 - a. Modify common application objects, such as schedules, calendars, and set points in a graphical manner.
 - 1. Schedule times will be adjusted using a graphical slider, without requiring any keyboard entry from the operator.
 - 2. Holidays shall be set by using a graphical calendar, without requiring any keyboard entry from the operator.
 - b. Commands to start and stop binary objects shall be done by right-clicking the selected object and selecting the appropriate command from the pop-up menu. No entry of text shall be required.
 - c. View logs and charts
 - d. View and acknowledge alarms
 - e. Setup and execute SQL queries on log and archive information
 - 6. The system shall provide the capability to specify a user’s (as determined by the log-on user identification) home page. Provide each specific user a defined home page based on

their usage requirements. From the home page, links to other views, or pages in the system shall be possible, if allowed by the system administrator.

7. Graphic screens on the Web Browser client shall support hypertext links to other locations on the Internet or on Intranet sites, by specifying the Uniform Resource Locator (URL) for the desired link.
8. BIPC, and NIPC Devices shall also serve an appropriate standard HTML5 graphic representative of the physical equipment being controller. The owner shall also be able to connect directly to these devices to see a local HTML page graphic of the equipment being controller. A HMTL Summary Page graphic shall also be provided for any MIPC from the Niagara Server.

1.21 SERVER FUNCTIONS AND HARDWARE

- A. An existing Vykon Enterprise Server is located on the City of Raleigh Wide Area Network. The server shall support all Network Area Controllers (NAC) connected to the customer's network whether local or remote. All programming, software upgrades and setup of the Enterprise Server shall be by Enterprise Developer.

1.22 COMPONENT LIBRARIES

- A. A standard library of components shall be included for development and setup of application logic, user interface displays, system services, and communication networks.
- B. The components in this library shall be capable of being copied and pasted into the user's database and shall be organized according to their function. In addition, the user shall have the capability to group components created in their application and store the new instances of these components in a user-defined library.
- C. In addition to the standard libraries specified here, the supplier of the system shall maintain an on-line accessible (over the Internet) library, available to all registered users to provide new or updated components and applications as they are developed.
- D. All control components shall conform to the control component specified in the BACnet specification.
- E. The component library shall include components to support the integration of devices connected to the Network Area Controller (NAC). At a minimum, provide the following as part of the standard library included with the programming software:
 1. For BACnet devices, provide the following components at a minimum:
 - a. Analog In
 - b. Analog Out
 - c. Analog Value
 - d. Binary

- e. Binary In
 - f. Binary Out
 - g. Binary Value
 - h. Multi-State In
 - i. Multi-State Out
 - j. Multi-State Value
 - k. Schedule Export
 - l. Calendar Export
 - m. Trend Export
 - n. Device
2. For each BACnet component, provide the ability to assign the component a BACnet device and component instance number.
 3. For BACnet devices, provide the following support at a minimum
 - a. Segmentation
 - b. Segmented Request
 - c. Segmented Response
 - d. Application Services
 - e. Read Property
 - f. Read Property Multiple
 - g. Write Property
 - h. Write Property Multiple
 - i. Confirmed Event Notification
 - j. Unconfirmed Event Notification
 - k. Acknowledge Alarm
 - l. Get Alarm Summary
 - m. Who-has
 - n. I-have
 - o. Who-is
 - p. I-am
 - q. Subscribe COV
 - r. Confirmed COV notification
 - s. Unconfirmed COV notification
 - t. Media Types
 - u. Ethernet
 - v. BACnet IP Annex J
 - w. MSTP
 - x. BACnet Broadcast Management Device (BBMD) function
 - y. Routing

1.23 BACNET NETWORK MANAGEMENT

- A. The Network Area Controller shall support the integration of device data from BACnet TCP/IP for all devices provided in Division 230900. If third-party equipment not provided in this section cannot be supplied as BACnet IP or supplied with a BACnet IP gateway, then FMCS Supplier shall provide the appropriate BACnet MSTP or Modbus driver to connect to the third-

party system device. The solution to connect the third-party device must be approved in advance by Owner's representative. (also see section 1.5.F.1)

- B. Provide the required components in the library, included with the Graphical User Interface programming software, to support the integration of the BACnet system data into the FMCS. Components provided shall include at a minimum:
 - 1. Read/Write BACnet AI Points
 - 2. Read/Write BACnet AO Points
 - 3. Read/Write BACnet AV Points
 - 4. Read/Write BACnet BI Points
 - 5. Read/Write BACnet BO Points
 - 6. Read/Write BACnet BV Points
- C. All scheduling, alarming, logging and global supervisory control functions, of the BACnet system devices, shall be performed by the Network Area Controller.
- D. The FMCS supplier shall provide a BACnet IP system communications driver. The equipment system vendor that provided the equipment utilizing BACnet shall provide documentation of the system's interface and shall provide factory support at no charge during system commissioning
- E. BACnet Conformance:
 - 1. Logic controllers shall as a minimum support MS/TP BACnet LAN type. They shall communicate directly via this BACnet LAN at 9.6, 19.2, 38.4 and 76.8 Kbps, as native BACnet devices. Logic controllers shall be of BACnet conformance class 3 and support all BACnet services necessary to provide the following BACnet functional groups:
 - a. Files Functional Group
 - b. Reinitialize Functional Group
 - c. Device Communications Functional Group
 - 2. Refer to Section 22.2, BACnet Functional Groups, in the BACnet Standard, for a complete list of the services that must be directly supported to provide each of the functional groups listed above. All proprietary services, if used in the system, shall be thoroughly documented and provided as part of the submittal data. All necessary tools shall be supplied for working with proprietary information.

1.24 BACNET IP COMMUNICATING THERMOSTATS

- A. The manufacturer of the Thermostat hardware and software components must be primarily engaged in the manufacture of BAS as specified herein and must have been so for a minimum of five (5) years.
- B. The manufacturer shall be ISO 9001:2000 certified. This is to ensure that all manufacturing, design and support policies comply with a minimum quality assurance standard. Corporate

quality assurance policies should be available for examination upon request by the owner or his agent.

- C. The manufacturer of the hardware and software components shall have a technical support group accessible via a toll-free number that is staffed with qualified personnel, capable of providing instruction and technical support service for networked control systems.
- D. Acceptable providers of the Communicating Thermostat hardware and software components as specified herein are as follows. Acceptance as a product provider does not provide approval to be an acceptable FMCS Systems Integrator.
 - 1. Contemporary Controls
 - 2. Viconics
 - 3. Owner-approved
- E. Communicating Thermostats shall be BACnet IP thermostats. FMCS Systems Integrator shall standardize on BACnet IP protocol for all thermostats.

1.25 BACNET IP VAV BOX CONTROLLERS (BIPC)

- A. The manufacturer of the hardware and software components must be primarily engaged in the manufacture of BAS as specified herein and must have been so for a minimum of five (5) years.
- B. The manufacturer shall be ISO 9001:2000 certified. Certification is to ensure that all manufacturing, design and support policies comply with a minimum quality assurance standard. Corporate quality assurance policies should be available for examination upon request by the owner or his agent.
- C. The manufacturer of the hardware and software components shall have a technical support group accessible via a toll-free number that is staffed with qualified personnel, capable of providing instruction and technical support service for networked control systems.
- D. Acceptable manufacturers of the hardware and software components as specified herein are as follows. Acceptance as a product manufacturer does not provide approval to be an acceptable Systems Integrator.
 - 1. Lynxspring JENEsys Edge VAV Niagara IP Controller
 - 2. Honeywell Optimizer VAV Controller
 - 3. Owner-approved

1.26 NETWORK IP CONTROLLER(S) STANDARDS

- A. Where beneficial, provide Plant, AHU, and VAV IP Controllers that can meet the required sequence of operation and can be custom programmed. All controllers shall be designed for easy installation and servicing including removable enclosures, removable terminals, and factory applied labels for all I/O. All internal points shall be fully supported by the Graphical User Interface (GUI), allowing the user to easily modify them and monitor them. All of the

internal programming points (e.g. variables, constants, PID's, timers, inputs and outputs) shall be exposed to the network on dedicated network variable outputs.

B. IP Equipment Control Units (Primary Systems such as Chiller, Boiler, Water System and a VAV AHU)

a. Acceptable Plant IP Controllers (Primary Systems such as Chiller, Boiler, Water Systems):

- i. Vykon Edge 10
- ii. Honeywell Optimizer Unitary Controller
- iii. Lynxspring JENEsys Edge 534 Niagara Controller
- iv. Owner-approved

b. General Requirements

- i. Communication Protocol: Supports IP-based protocols such as BACnet/IP, BACnet MSTP, and Modbus TCP/IP.
- ii. Network Interface: Ethernet (10/100/1000 Mbps) with 2 RJ45 connectors; optional support for Wi-Fi or fiber optic connections.
- iii. Controller Type: Designed for plant control applications (managing and controlling various mechanical and electrical plant equipment, such as boilers, chillers, pumps, and cooling towers).
- iv. Power Supply: 24 VAC/VDC or 120/240 VAC, with power consumption specifications (e.g., 15 VA typical).

c. Inputs/Outputs

i. **Analog Inputs (AI):**

- a) Number: 8-16 AI
- b) Type: 0-10V, 4-20mA, thermocouple, RTD, thermistor
- c) Resolution: 12-bit or higher
- d) Accuracy: $\pm 0.1\%$ of full scale

ii. **Digital Inputs (DI):**

- a) Number: 8-16 DI
- b) Type: Dry contact, pulse inputs, or voltage inputs (24VDC typical)
- c) Minimum pulse width: 1 ms

iii. **Analog Outputs (AO):**

- a) Number: 4-8 AO
- b) Type: 0-10V, 4-20mA
- c) Resolution: 12-bit or higher
- d) Accuracy: $\pm 0.1\%$ of full scale

iv. **Digital Outputs (DO):**

- a) Number: 8-16 DO
- b) Type: Relay outputs, 24V AC/DC, or dry contact
- c) Maximum current: 3A per relay

d. **Control Features**

- i. Control Algorithms: Supports advanced control algorithms, including PID, cascade, and sequence control for complex plant operations.
- ii. Setpoint Management: Configurable setpoints for temperature, pressure, flow, and other key parameters.
- iii. Scheduling: Integrated scheduling for daily, weekly, and holiday operation, with override capabilities.
- iv. Trending and Logging: Real-time and historical data logging with configurable sampling rates; supports trend graphs and data export.
- v. Alarm Management: Built-in alarm handling with email/SMS notifications; configurable alarm thresholds and priorities.

e. **Software and Programming**

- i. Programming Language: Support for graphical programming (e.g., ladder logic, block programming).
- ii. Firmware: Field upgradable via IP network; support for secure firmware updates.
- iii. Configuration Interface: Web-based interface, local display, or PC-based software for configuration, monitoring, and diagnostics.

f. **Integration**

- i. Seamless integration with Building Management Systems (BMS) using Niagara FOXs and other supervisory control systems using open protocols.
- ii. Integrate with the building's BMS for centralized monitoring and control of all energy-related systems within the facility

g. Security

- i. Multi-level user authentication, SSL/TLS encryption, and secure boot mechanisms.

h. Environmental Specifications

- i. Operating Temperature: 0°C to 50°C (32°F to 122°F), extended temperature range options available.
- ii. Storage Temperature: 20°C to 70°C (-4°F to 158°F)
- iii. Humidity: 10% to 90% RH non-condensing

i. Ingress Protection (IP) Rating

- i. IP20 or better; optional IP54 or higher for industrial environments.

j. Integration and Interoperability

- i. Third-Party Device Support: Compatibility with a wide range of third-party sensors, actuators, and equipment.
- ii. Interoperability: Supports interoperability with existing plant systems and devices, ensuring flexibility in system upgrades and expansions.

k. Additional Features

- i. Remote Access: Secure remote access via web interface or mobile app, with support for VPN and encrypted connections.
- ii. Redundancy: Support for controller redundancy and failover mechanisms to ensure continuous operation.

l. Physical Characteristics

- i. Mounting: Suitable for DIN rail, panel, or wall mounting, with optional enclosures for harsh environments.

m. IP Plant Controllers shall include:

- i. NXP iMX6 SoloX2: 800 MHz ARM Cortex-A9/M4 or equal
- ii. 512 MB DDR SDRAM
- iii. 2GB total eMMC flash storage with user space set at 1GB
- iv. Powered from 24VAC/DC source
- v. 5 Universal inputs: Type 3 (10K) thermistors, 0-100K ohm, 0-10VDC, 0-20mA with external resistor, Dry Contact

- vi. 2 Analog outputs: 0-10VDC, 4mA max output current
 - vii. 3 Digital outputs: Triac, 24VAC @.5 amp
 - viii. 2 10/100MB Ethernet ports capable of daisy chaining
 - ix. 1 RS-485 serial port
 - n. VAV IP Controller Products:
 - i. Lynxspring JENEsys Edge VAV Niagara Controller
 - ii. Honeywell Optimizer VAV Controller
- a) **General Requirements**
 - a. **Communication Protocol:** Supports IP-based communication protocols such as BACnet/IP or BACnet/MSTP.
 - b. **Network Interface:** 2 - Ethernet (10/100 Mbps) with RJ45 connector, optional support for Wi-Fi.
 - c. **Controller Type:** Programmable controller designed specifically for VAV applications.
 - d. **Power Supply:** 24 VAC/VDC
 - e. **Serial Communications:** 2 – RS485 ports
 - b) **Environmental Specifications**
 - a. **Operating Temperature:** 0°C to 50°C (32°F to 122°F)
 - b. **Storage Temperature:** -20°C to 70°C (-4°F to 158°F)
 - c. **Humidity:** 10% to 90% RH non-condensing
 - d. Compact design for small enclosures
 - e. **Mechanical Protection: Ingress Protection (IP) Rating:** IP20 or better (protection against solid objects larger than 12.5 mm and no protection against water)
 - f. **Vibration Resistance:** Meets IEC 60068-2-6 standards for vibration, ensuring durability in environments with mechanical vibrations.
 - g. **Shock Resistance:** Meets IEC 60068-2-27 standards for shock, ensuring the controller can withstand sudden impacts.
 - h. **RoHS (Restriction of Hazardous Substances):** Compliant with RoHS directives, ensuring the controller is free from hazardous materials like lead, mercury, and cadmium.

- i. REACH (Registration, Evaluation, Authorization, and Restriction of Chemicals): Compliant with REACH regulations, ensuring the use of safe chemicals in the controller's manufacturing process.
- j. Electromagnetic Emissions: Complies with EN 55022 (Class B) or equivalent standards for emission limits.
- k. Electromagnetic Immunity: Complies with EN 55024 or equivalent standards, ensuring immunity to electromagnetic interference (EMI).
- l. Overvoltage Category: Overvoltage Category II (suitable for equipment connected to the mains, including standard electrical appliances and portable tools).
- m. Chemical Resistance: Resistant to common chemicals found in HVAC environments, such as cleaning agents, lubricants, and airborne pollutants.
- n. Material: Enclosure made of flame-retardant, UV-resistant, and corrosion-resistant plastic or metal, suitable for indoor use.
- o. Mounting: Suitable for DIN rail or panel mounting, with options for wall mounting in harsher environments.

c) **Inputs/Outputs**

- **Analog Inputs (AI):**

- Number: 4-6 AI
- Type: 0-10V, 4-20mA, thermistor, resistance inputs
- Resolution: 12-bit or higher
- Accuracy: $\pm 0.1\%$ of full scale

- **Digital Inputs (DI):**

- Number: 4-8 DI
- Type: Dry contact or pulse inputs
- Minimum pulse width: 10 ms

- **Analog Outputs (AO):**

- Number: 2-4 AO
- Type: 0-10V, 4-20mA
- Resolution: 12-bit or higher
- Accuracy: $\pm 0.1\%$ of full scale

- **Digital Outputs (DO):**

- Number: 4-8 DO
- Type: Relay outputs, 24V AC/DC, or dry contact
- Maximum current: 2A per relay

d) **Control Features**

- a. **Control Algorithms:** Configurable for pressure-independent and pressure-dependent VAV applications.
- b. **Airflow Measurement:** Integrated airflow sensors or support for external airflow transducers.
- c. **Damper Control:** Proportional, integral, and derivative (PID) control for damper positioning.
- d. **Fan Control:** Support for single-speed, multi-speed, or variable speed fan control.
- e. **Temperature Control:** Built-in support for heating and cooling control loops.
- f. **Occupancy Control:** Support for occupancy sensors for energy-saving mode.

e) **Software and Programming**

- a. **Programming Language:** Support for graphical programming (e.g., ladder logic, block programming).
- b. **Firmware:** Field upgradable via IP network.
- c. **Configuration:** Web-based configuration interface, local display, or PC-based software.
- d. **Scheduling:** Built-in scheduler with support for weekly schedules, holiday schedules, and exceptions.
- e. Ready to use sequence of operations.

f) **Integration**

- a. **BMS Compatibility:** Seamless integration with building management systems (BMS) using Niagara FOXs.
- b. **Third-Party Device Support:** Ability to integrate with third-party sensors, actuators, and other devices.

g) **Compliance and Certifications**

- a. **Standards:** Compliant with ASHRAE, UL, CE, and FCC standards as applicable.

h) **Additional Features**

- a. **Logging and Alarming:** Built-in data logging with adjustable intervals, event logging, and alarm notifications.
- b. **User Interface:** Optional local display with touchscreen or button-based interface for on-site configuration.
- c. **Remote Access:** Support for remote access via secure web interface IP-VAV-CTRL shall be 32-bit microprocessor-based operating at a minimum of 400 MHz.
- i) They shall be multi-tasking, real-time digital control processors consisting of modular hardware with plug-in enclosed processors, communication controllers, power supplies and input/output point modules.
- j) Controller size shall be sufficient to fully meet the requirements of this specification and the attached point list.
- k) Each IP-VAV-CTRL shall have minimum of 512MB memory, with a minimum of 1GB non-volatile flash, to support its own operating system and databases, including:
 - a. Control processes.
 - b. Maintenance support applications
 - c. Custom processes
 - d. The IP-VAV-CTRL shall have a Real-time clock, a supercapacitor for 24-hour data retention or Battery-powered clock to store setup values (Y:M:D:H:M:S)
- l) 2 x USB 2.0 Expansion ports for:
 - a. 802.11 Wi-Fi Adapter enabling wireless connectivity including:
 - 1. 'Hot Spot'
 - 2. Client
 - 3. Access Point
 - b. If the above functionality is not available, then appropriate wireless router(s) and switches must be supplied to provide the functionality.
 - c. shall contain a "FIPS 140-2 Inside" cryptographic module

iii. Variable Air Volume (VAV) Terminal Control Units (TCU)

- a) The VAV TCU controllers shall be powered from a 24 VAC source and shall function normally under an operating range of 20 to 28 VAC ($\pm 15\%$), allowing for power source fluctuations and voltage drops.

- b) The BAS contractor shall provide a dedicated power source and separate isolation transformer for each controller unable to function normally under the specified operating range.
- c) The controllers shall also function normally under ambient conditions of 32 °F to 122 °F and 5% to 90% RH (non-condensing).
- d) Provide each controller with a suitable cover or enclosure to protect the intelligence board assembly.
- e) The VAV TCU shall include a built-in 'flow thru' differential pressure transducer.
 - a. The controller shall convert this value to actual air flow.
 - b. Single point differential pressure sensing device is not acceptable.
 - c. Membrane based pressure differential transducer is not acceptable.
 - d. The VAV TCU differential pressure transducer shall have a measurement range of 0 to 2 in. W.C. and measurement accuracy of $\pm 4\%$ at 0.05 to 2 in. W.C. and a minimum resolution of 0.0001 in. W.C., insuring primary air flow conditions shall be controlled and maintained to within $\pm 5\%$ of setpoint at the specified minimum and maximum air flow parameters.
 - e. VAV TCU differential pressure transducer requiring periodic zero value air flow calibration is not acceptable.
- f) The BAS contractor shall verify the type of differential pressure sensors used in the existing boxes and ensure compatibility with the VAV TCU controllers.
- g) The VAV TCU shall include provision for air flow balancing using a local air flow balancing interface.
- h) An Intelligent Space Sensor (ISS) shall be used for balancing air flow.
 - a. In lieu of an ISS, a portable air flow balancing interface capable of balancing air flow is acceptable.
- i) The portable air flow balancing interface shall connect to the VAV TCU or the matching room temperature sensor.
- j) The VAV TCU shall also provide an air flow balancing tool.
- k) This tool shall allow the air balancer to manually control the action of the actuator including the following function: open VAV damper, close VAV damper, open all VAV dampers, and close all VAV dampers.

- l) Systems not able to provide a web based air balance tool or a portable air flow balancing interface or an Intelligent Space Sensor (ISS) capable of balancing air flow as part of the VAV TCU controller shall provide an individual full time technician during the air flow balancing process to assure full balance compliance.
- m) The VAV box controller shall interface to a matching room temperature sensor as previously specified. The controller shall function to maintain space temperature to within ± 1.5 °F of setpoint at the room sensor location.
- n) Each controller shall also incorporate an algorithm that allows for resetting of the associated air handling unit discharge temperature if required to satisfy space requirements.
 - a. This algorithm shall function to signal the respective controller to perform the required discharge temperature reset in order to maintain space temperature setpoint.
- o) It shall be possible to view and reset the space temperature, temperature setpoint, maximum airflow setting, minimum airflow setting, and actual airflow, through the BAS LAN.

1.27 MODBUS SYSTEM INTEGRATION

- A. The Network Area Controller shall support the integration of device data from Modbus RTU, Ascii, or TCP control system devices. The connection to the Modbus system shall be via an RS-232, RS485, or Ethernet IP as required by the device.
- B. Provide the required components in the library, included with the Graphical User Interface programming software, to support the integration of the Modbus system data into the FMCS. Components provided shall include at a minimum:
 - 1. Read/Write Modbus AI Registers
 - 2. Read/Write Modbus AO Registers
 - 3. Read/Write Modbus BI Registers
 - 4. Read/Write Modbus BO Registers
- C. All scheduling, alarming, logging and global supervisory control functions, of the Modbus system devices, shall be performed by the Network Area Controller.
- D. The FMCS supplier shall provide a Modbus system communications driver. The equipment system vendor that provided the equipment utilizing Modbus shall provide documentation of the system's Modbus interface and shall provide factory support at no charge during system commissioning
- E. Provide a Modbus Interface to the following equipment:
 - 1. switchgear
 - 2. packaged pumping system

3. building energy metering

1.28 THIRD PARTY INTEGRATION

- A. The Network Area Controller shall support the integration of device data from the existing control system. The connection to the existing system shall be via an RS-232 connection between the Network Area Controller and the existing control system {if applicable on this project}.
- B. Provide the required data points from the third-party integration per sequence of operations and/or City of Raleigh points list

2.9 SENSORS

- A. All control items, except thermostats, sensors and transmitters located in rooms shall be properly identified with engraved plastic nameplates permanently attached. Nameplates shall have white letters on a black background.
- B. All sensors shall be provided in NEMA 4X enclosures where exposed to the Pool environment.
- C. Room thermostat, sensor and transmitter locations shall be coordinated to align vertically or horizontally with adjacent light switches or other control devices. Room thermostats and sensors shall be mounted with the bottom 5'-0" above the floor. Sensors installed in areas where they are subject to physical abuse (ex: gymnasiums) shall be furnished with protective type aspirating guards. Sensors installed on exterior walls shall be installed on non-conductive (cork) sub-base. Sensors shall have plus or minus local control feature.
- D. Temperature Sensors: Thermistor type with an accuracy of plus or minus 0.40 degree F over the entire control range. Sensors for pipe installations shall be immersion type, brass well, and thermistor with integral lead wire. Sensors for duct application shall be insertion probe type, stainless steel probe, integral handi-box, and thermistor with integral lead wire. Space temperature sensors shall be compatible with the unit controller and shall be provided in a decorative metal or plastic enclosure (NEMA 4X where exposed to pool environment). Space temperature sensors shall be provided with setpoint and temperature indication only. Outdoor temperature sensors shall be mounted inside a protective weather and sun shield and shall be located on a North wall.
- E. Humidity Sensors: Thin-film capacitive type sensor with on-board nonvolatile memory, accuracy to plus or minus two percent (2%), 12 - 30 VDC input voltage, analog output (0 - 10 VDC). Operating range shall be 5 to 95% RH and -40 to 170 degree F. Duct mounted type sensors shall have a stainless steel insertion element, sealed to prohibit corrosion. Sensors shall be selected for wall, duct or outdoor type installation as appropriate.
- F. Carbon Dioxide Sensors (CO2): Sensors shall utilize Non-dispersive infrared technology (N.D.I.R.), repeatable to plus or minus 20 PPM. Sensor range shall be 0 - 2000 PPM. Accuracy shall be plus or minus five percent (5%) or 50 PPM, whichever is greater. Response shall be less than one minute. Input voltage shall be 20 to 30 VAC/DC. Output shall be 0 - 10 VDC. Sensor shall be wall or duct mounted type, as appropriate for the application, housed in a high impact plastic enclosure.

- G. Differential Air Pressure Switch: Differential pressure switches for proving fan operation or sense dirty air filters shall be SPDT type, UL approved, and selected for the appropriate operating range of the equipment to which it is applied. Sensor shall have ¼" compression type fittings and shall have an adjustable setpoint. Furnish with ¼" barbed type static pressure tips.
- H. Current Switches (Type 1): For proving fan or pump operational status, provide solid or split-core type current status switches with adjustable setpoint and solid-state internal circuitry. Current switch shall have induced power, trip point set adjustment to plus or minus 1% over a range of 1 to 135 amps, trip and power LED, and field adjustable to indicate both On-Off conditions and loss of load (broken belt, etc.). Units shall have a five-year manufacturer's warranty. Current switches shall be Hawkeye Series H-908 by Veris Industries, or approved equal.
- I. Current Switches (Type 2): For proving fan or pump operational status, provide solid or split-core type current switches ("Go/No" type). Current switch shall have induced power, 100 percent solid state with no moving parts. Units shall have a five-year manufacturer's warranty. Current switches shall be Hawkeye series H-900 by Veris Industries, or approved equal.
- J. Low Temperature Sensors: For sensing low temperatures in air handling units, provide SPST type switch, 35 to 45 degree F range, manual reset, vapor charged twenty foot long sensing element, and 120-volt electrical power connection. Low temperature sensor ("freeze-stat") shall be JCI Model A11A-1, or equal.
- K. Pressure Transmitters: For sensing static pressure in a duct system (usually for VAV systems), provide a pressure transmitter with integral capacitance type sensing action, solid state circuitry, accuracy of plus or minus 1% of range, zero and span adjustments, 10 to 35 VDC operating voltage, 4 to 20mA output, and integral inlet port connections. Select pressure range suitable for the application. Differential pressure transmitter shall be Ashcroft CXLdp, or approved equal.
- L. Line Voltage Thermostats: For control of equipment using line voltage on-off thermostats (exhaust fans, unit heaters, etc.) provide 120 volt UL Listed wall mounted thermostats. Thermostat shall have a range of 50 to 90 degree F with minimum 2 degree F differential, snap acting switch, and dial adjustment for temperature setting. Line Voltage Thermostats shall be Honeywell series T631 series or approved equal.
- M. Firestat: For sensing sudden increases in duct temperature (ex: fire condition), provide 120 volt UL Listed SPST switch with adjustable setpoint that breaks the circuit on a rise in temperature above the setpoint and de-energizes the air handling unit fan.
- N. Aquastat: For sensing temperature of a fluid within a pipe system, provide 120-volt SPST strap-on type aquastat, temperature control range of 65 to 200-degree F (adjustable). Aquastat shall be HW Model L6006C1018, or equal.
- O. Air Flow Monitoring Device
 - 1. Basis-of-Design Product: Subject to compliance with requirements, provide Ebtron Advantage Gold Series for use with GTx116 transmitters or a comparable product by one of the following:
 - a. Tek-Air Systems or equal

2. Provide airflow/temperature measurement devices (ATMD) where indicated on the plans. Fan inlet measurement devices shall not be substituted for duct or plenum measurement devices indicated on the plans.
3. Each ATMD shall consist of one or more sensor probes and a single, remotely mounted, microprocessor-based transmitter capable of independently processing up to 16 independently wired sensor assemblies.
 - a. Each sensor assembly shall contain two individually wired, hermetically sealed bead-in-glass thermistors.
 - b. Thermistors shall be mounted in the sensor assembly using a marine-grade, waterproof epoxy. Thermistor leads shall be protected and not exposed to the environment.
 - c. The airflow rate of each sensor assembly shall be equally weighted and averaged by the transmitter prior to output.
 - d. The temperature of each sensor assembly shall be velocity weighted and averaged by the transmitter prior to output.
 - e. Each transmitter shall have a 16-character alpha-numeric display capable of displaying airflow, temperature, system status, configuration settings and diagnostics.
 - f. Devices using chip-in-glass or diode-case chip thermistors are not acceptable.
 - g. Devices using less than two thermistors in each sensor assembly are not acceptable.
 - h. Devices using platinum wire RTDs are not acceptable.
 - i. Devices having electronic circuitry mounted in or at the sensor probe are not acceptable.
 - j. Pitot tubes and arrays are not acceptable.
 - k. Vortex shedding devices are not acceptable.
4. All Sensor Probes
 - a. Each sensor assembly shall independently determine the airflow rate and temperature at each measurement point.
 - b. Each sensor assembly shall be calibrated at a minimum of 16 airflow rates and 3 temperatures to standards that are traceable to the National Institute of Standards and Technology (NIST).
 - c. Airflow accuracy shall be +/-2% of Reading over the entire operating airflow range.
 - i. Devices whose accuracy is the combined accuracy of the transmitter and sensor probes must demonstrate that the total accuracy meets the performance requirements of this specification throughout the measurement range.
 - d. Temperature accuracy shall be +/-0.15° F over the entire operating temperature range of -20° F to 160° F.
 - e. The operating humidity range for each sensor probe shall be 0-99% RH (non-condensing).
 - f. Each sensor probe shall have an integral, U.L. listed, plenum rated cable and terminal plug for connection to the remotely mounted transmitter. All terminal plug interconnecting pins shall be gold plated.
 - g. Each sensor assembly shall not require matching to the transmitter in the field.

- h. A single manufacturer shall provide both the airflow/temperature measuring probe(s) and transmitter for each measurement location.

5. Duct and Probes

- a. Probes shall be constructed of extruded, gold anodized, 6063 aluminum tube. All wires within the aluminum tube shall be Kynar coated.
- b. Probe assembly mounting brackets shall be constructed of 304 stainless steel. Probe assemblies shall be mounted using one of the following options:
 - i. Insertion mounted through the side or top of the duct
 - ii. Internally mounted inside the duct or plenum
 - iii. Standoff mounted inside the plenum
- c. The number of sensor housings provided for each location shall be as follows:

Duct Area (sq.ft.)	Total # Sensors / Location
<2	4
2 to < 4	6
4 to < 8	8
8 to <16	12
>=16	16

- d. The operating airflow range shall be 0 to 5,000 FPM unless otherwise indicated on the plans.

6. Fan Inlet Probes

- a. Sensor assemblies shall be mounted on 304 stainless steel housings.
- b. Mounting rods shall be field adjustable to fit the fan inlet and constructed of nickel plated steel.
- c. Mounting feet shall be constructed of 304 stainless steel.
- d. The operating airflow range shall be 0 to 10,000 FPM unless otherwise indicated on the plans.

7. Transmitters

- a. The transmitter shall have an integral LCD display capable of simultaneously displaying airflow and temperature. The LCD display shall be capable of displaying individual airflow and temperature readings of each independent sensor assembly.
- b. The transmitter shall be capable of field configuration and diagnostics using an on-board pushbutton interface and LCD display.
- c. The transmitter shall have a power switch and operate on 24 VAC (isolation not required).
 - i. The transmitter shall use a switching power supply fused and protected from transients and power surges.

- ii. The transmitter shall use “watch-dog” circuitry to assure reset after power disruption, transients and brown-outs.
 - d. All interconnecting pins, headers and connections on the main circuit board, option cards and cable receptacles shall be gold plated.
 - e. The operating temperature range for the transmitter shall be -20° F to 120° F. The transmitter shall be installed at a location that is protected from weather and water.
 - f. The transmitter shall be capable of communicating with other devices using the following interface option: Linear analog output signals for airflow and temperature: Field selectable, fuse protected and isolated, 0-10VDC/4-20mA (4-wire)
8. The transmitter shall be capable of accepting an infra-red interface card for downloading airflow and temperature data or uploading transmitter configuration data using a handheld PDA (Palm or Microsoft Windows Mobile operating systems).
- a. Provide PDA upload/download software.
 - i. Download software shall be capable of displaying and saving individual sensor airflow rates, the average airflow rate, individual sensor temperatures and the average temperature received from the transmitter.
 - ii. Upload software shall be capable of displaying and saving all setup parameters that can be configured using the on-board pushbutton interface and LCD display.
9. The ATMD shall be UL listed as an entire assembly.
10. The ATMD shall carry the CE Mark for European Union shipments.
11. The manufacturer’s authorized representative shall review and approve placement and operating airflow rates for each measurement location indicated on the plans.

1.29 DAMPERS AND ACTUATORS

- A. Damper actuators shall be sized by the Systems Integrator for the intended application. Unless noted otherwise, dampers will be furnished by the Systems Integrator for all field installed dampers that are not included as part of the equipment. In general, provide opposed blade type dampers for modulating control and parallel type dampers for two-position control applications. Actuators shall be Honeywell MS or ML series actuators, Belimo LMB24-3-T, or similar actuators with the City of Raleigh's approval,
- B. Control Dampers: When indicated to be furnished by the Systems Integrator, control dampers shall be equal to Ruskin CD30VG2 or Honeywell D2 or D3 series dampers. Provide all automatic control dampers not specified to be integral with other equipment. Frames shall be 5 inches wide and of no less than 16-gauge galvanized steel. Inter-blade linkage shall be within the frame and out of the air stream. Blades shall not be over 8 inches wide nor less than 16-gauge galvanized steel triple V type for rigidity. Bearings shall be acetal, oilite, nylon or ball-bearing with ½ inch diameter plated steel shafts. Dampers shall be suitable for temperature ranges of -40 to 180F. All proportional control dampers shall be opposed or parallel blade type

as hereinafter specified and all two-position dampers shall be parallel blade types. Dampers shall be sized to meet flow requirements of the application. The sheet metal contractor shall furnish and install baffles to fit the damper to duct size. Baffles shall not exceed 6". Dampers with dimensions of 24 inches and less shall be rated for 3,000 fpm velocity and shall withstand a maximum system pressure of 5.0 in. wc. Dampers with dimensions of 36 inches and less shall be rated for 2,500 fpm velocity and shall withstand a maximum system pressure of 4.0 in. wc. Dampers with dimensions of 48 inches and less shall be rated for 2,000 fpm velocity and shall withstand a maximum system pressure of 2.5 in. wc. Side seals shall be stainless steel of the tight-seal spring type. Dampers shall be minimum leakage type to conserve energy and the temperature control manufacturer shall submit leakage data for all low leakage control dampers with the temperature control submittal. Maximum leakage for low leakage dampers in excess of sixteen inches square shall be 8 CFM per square foot at static pressure of 1 inch of WC. Low leakage damper blade edges shall be fitted with replaceable, snap-on, inflatable seals to limit damper leakage. Testing and ratings shall be in accordance with AMCA Standard 500. Damper blade width shall be no greater than 8 inches, and dampers over 48 inches wide by 74 inches high shall be sectionalized. Testing and ratings to be in accordance with AMCA Standard 500.

- C. Damper Actuators: Damper actuators shall be provided for all automatic dampers. Damper actuators controlled through the DDC system shall be low voltage electronic type, either modulating or two-position, as required to achieve the intended sequence of operation. Provide with spring return when required for fail-safe operation. Modulating dampers shall be positive positioning in response to a 2 - 10 VDC or 4 - 20mA control signal. Actuator shall include the capability of adding auxiliary switches for position indication. Furnish actuators other than spring return type with a release button (clutch) or handle on the actuator to allow for manual override. Power supply to the actuator shall be by 120 VAC, 24 VAC, or 24 VDC and the actuator shall be furnished with a factory installed 3-foot cable with end fitting for field connection. All actuators shall be UL Listed by the manufacturer. Actuators shall be Honeywell MS or ML series actuators.

1.30 CONTROL VALVES

- A. Control Valves: (Globe Type) Valves shall be Honeywell or equivalent. Control valves shall be 2-way or 3-way pattern as shown constructed for tight shutoff and shall operate satisfactory against system pressures and differentials. Two-position valves shall be 'line' size. Proportional control valves shall be sized for a maximum pressure drop of 5.0 psi at rated flow (except as may be noted on the drawings). Two-way water valves shall have equal percentage flow characteristics and three-way valves shall have equal percentage flow characteristics straight through and linear through the bypass. Provide valve position indicator on all valves. Leakage rate shall be no more than 0.05% of Cv.
 - 1. Valves 1/2 inch through 1 1/2 inch shall be screwed pattern except where solder connections are specified for valves 1/2 or 3/4 inches. Three-way valves bypass port shall be of one size reduced Cv to preclude the need for a bypass port balancing valve. Valve and cartridge replacement tool shall be configured for maintenance or replacement without draining the coil to prevent water spill; however, an integral isolation valve on the control valve outlet will also be acceptable. Valves shall close off against 58 psi minimum.

2. Two-inch valves shall be “screwed” configuration and 2-1/2 inch and larger valves shall be “flanged” configuration and ANSI-rated to withstand the pressures and temperatures encountered. Valves shall have stainless-steel stems and spring-loaded Teflon packaging with replaceable discs.
- B. Control Valves: (Characterized Ball Valves) Valves shall be Honeywell or equivalent. Control valves 1/2 to 2 inches shall be 2-way or 3-way forged brass screwed pattern as shown constructed for tight shutoff and shall operate satisfactory against system pressures and differentials. Two-position valves shall be ‘line’ size. Proportional control valves shall be sized for a maximum pressure drop of 5.0 psi at rated flow (except as may be noted on the drawings). Two-way water valves shall have equal percentage flow characteristics and three-way valves shall have equal percentage flow characteristics straight through and linear flow through the bypass. Leakage rate shall be ANSI Class IV (no more than 0.01% of Cv). Valves shall be rated for no less than 350 psig at no less than 250 degrees F. Provide a removable handle to operate valves manually during actuator power loss or failure.
 - C. Two-way valves shall close off against 100 psi minimum, and three-way valves shall close off against 40 psi minimum. Valves shall have stainless-steel or chemically nickel-plated brass stem and throttling port. Valves shall be tagged with Cv rating and model number.
 - D. Butterfly Control Valves: Valves shall be Honeywell or equivalent. Where specified butterfly control valves over 2” in size shall be cast iron body type for 2-way or 3-way applications specified constructed for tight shutoff and shall operate satisfactory against system pressures and differentials. Valves shall have tapped lugs for standard flange connection and designed for isolation and removal of downstream piping at full rated pressure. Two-position valves shall be ‘line’ size. Proportional control valves shall be sized for a maximum pressure drop of 5.0 psi at rated flow (except as may be noted on the drawings). Valves shall be rated for bubble tight shutoff at no less than 150 psi. Valve disc shall be aluminum bronze. Valve stems shall be stainless steel, with inboard top and bottom bronze bearings, and an external corrosion resistant top bearing to absorb actuator side thrust.

1.31 GREEN ROAD SUPPLEMENTAL SPECIFICATIONS

A. INTRODUCTION

This section provides additional specifications for the HVAC replacement project at Green Road Community Center.

B. Site Survey

The contractor shall conduct a comprehensive on-site survey to verify existing conditions for COR furnished network switches. Ensuring that there are sufficient data ports and infrastructure to support all new IP based controllers. On-site survey shall be conducted with contractor, City of Raleigh IT department, and a member of the COR IFS team.

C. Control Panels

The contractor shall utilize the existing control panels for the HVAC replacement project at Green Road CC.

D. JACE Controller

1. The contractor shall replace the existing JACE 8000 with a JACE 9000. The contractor shall adhere to Sections 1.2B and 1.4 for JACE controller specifications.
2. The contractor shall provide the latest approved Niagara 4.15 software release installed on all applicable controllers and workstations. Verify that all required updates, patches, and licensing requirements are current prior to project completion.

E. BACnet Communication

The contractor shall install the BACnet/IP communication protocol. BACnet MSTP shall not be used for any BACnet controllers.

F. Niagara Network

1. Station User

Configure a dedicated station user account that will be utilized to establish communication between Supervisor and controller client connections for the purpose of securely transmitting data between systems. User account will be configured to have admin privileges. Ensure that the station user is established and configured in full compliance with the owner's established standards and requirements. Coordinate directly with a member of the COR IFS team to confirm that the client connection user has been set up correctly and meets all specified criteria.

2. Client Connection

Configure the client connection on the Niagara Network between the Supervisor and controllers in full compliance with the owner's established standards and requirements. Coordinate directly with a member of the COR IFS team to confirm that the client connection has been set up correctly and meets all specified criteria.

- i. Configure the client connection on the Niagara Network between the controller and supervisor to utilize secure Foxs communication.
- ii. Configure the client connection on the Niagara Network to utilize a specified port number given by a member of the COR IFS team.
- iii. Configure the client connection on the Niagara Network to utilize a specified alarm class given by a member of the COR IFS team.

G. JACE Platform Configuration

1. Platform Passphrase

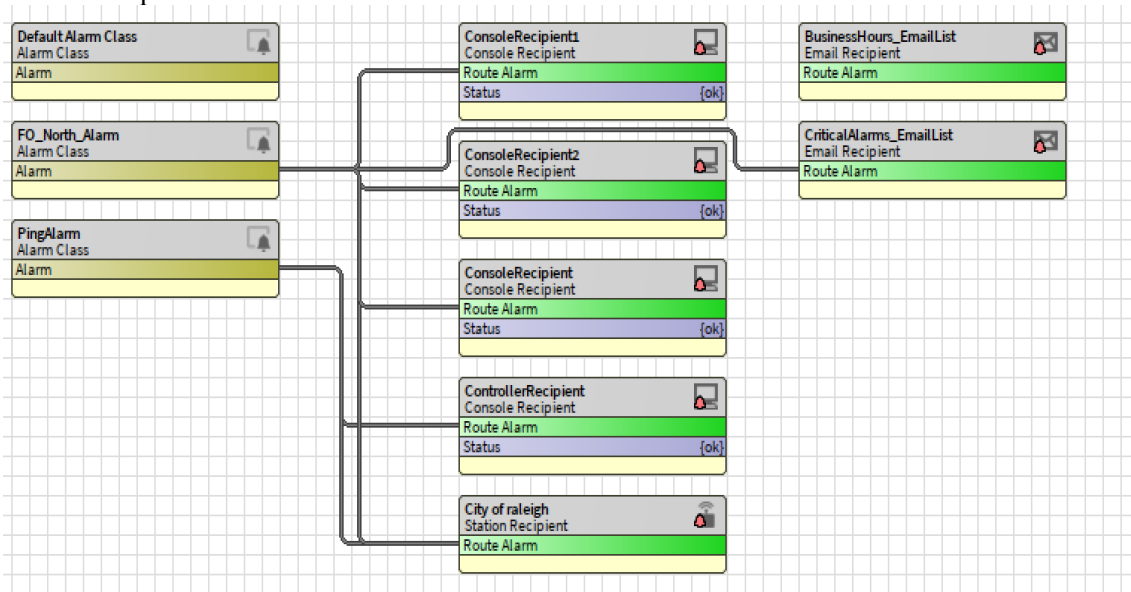
Ensure that the platform passphrase is established and configured in full compliance with the owner's established standards and requirements. Coordinate directly with a member of the COR IFS team to confirm that the passphrase has been set up correctly and meets all specified criteria.

2. Platform User

Configure a dedicated platform user account that will be utilized to establish communication between Supervisor and controller platforms for the purpose of securely transmitting data between systems. Ensure that the platform user is established and configured in full compliance with the owner's established standards and requirements. Coordinate directly with a member of the COR IFS team to confirm that the platform user has been set up correctly and meets all specified criteria.

H. Alarm Class and Routing

The Contractor shall configure all BACnet Ping fail alarms using the 'PingAlarm' Alarm_Class, while all other alarms shall use 'FO_North_Alarm'. All alarms must be routed to a local alarm console on the Niagara Station and to the City_of_Raleigh Supervisor Station using StationRecipient.



I. POINTS, TAGGING, TRENDS AND ALARM

The Facility Management and Control System (FMCS) contractor is responsible for implementing the Building Automation requirements for all point lists specified in 02-230900 – Instrumentation and Control Point List Requirements. If a type of equipment is missing from 02-230900 – Instrumentation and Control Point List Requirements, the FMCS contractor is responsible for providing a point list using the table format shown in the example below. The FMCS contractor will coordinate approval of the point list with the owner's Building Automation Specialist.

Point List Table Example

Description	Point Name	Tags	Point Type	UoM	History	Override
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The FMCS contractor is responsible for providing an alarm list for all equipment in the project under the control of the Facility Management and Control System (FMCS). The FMCS contractor will coordinate approval of the alarm list with the owner's Building Automation Specialist. The FMCS contractor will maintain the current alarm list at the site.

Package Heating Cooling Point List Requirements

Description	Point Name	Tags	Point Type	Override	UoM	History
DA Temp	DaTemp	discharge, air, temp, sensor	AI		F	Int, 10min
Mixed Air Temp	MaTemp	mixed, air, temp, sensor	AI		F	Int, 10min
Zone Temperature	ZoneTemp	zone, air, temp, effective, sp	AI		F	Int, 10min
Zone Temperature Setpoint	ZoneTempSp	zone, air, temp, effective, sp	AV	x	F	COV
Occupied Cooling Setpoint	OccCoolSp	occ, cooling, zone, air, temp, sp, hisMode:cov	AV	x	F	COV
Occupied Heating Setpoint	OccHeatSp	occ, heating, zone, air, temp, sp, hisMode:cov	AV	x	F	COV
Unoccupied Cooling Setpoint	UnoccCoolSp	unocc, cooling, zone, air, temp, sp, hisMode:cov	AV	x	F	COV
Unoccupied Heating Setpoint	UnoccHeatSp	unocc, heating, zone, air, temp, sp, hisMode:cov	AV	x	F	COV
Outside Air Damper Command	OaDmprCmd	outside, air, damper, cmd	BO	x	Open/Closed	COV
Outside Air Damper Position	OaDmprPos	outside, air, damper, cmd	AI	x	%	Int, 10min
Zone Humidity	ZoneHumidity	air, humidity, sensor	AI		%	Int, 10min
Dehum Setpoint	DehumSp	air, humidity, sp	AV	x	%	COV
Compressor Command	CompCmd	runCmd, compressor, equip	BO	x	On/Off	COV
Compressor Status	CompSts	runStatus, stage, compressor, equip	BI		On/Off	COV
Fan Command	FanCmd	fan, run, cmd	BO	x	On/Off	COV
Fan Status	FanSts	fan, run, sensor	BI		On/Off	COV
Fan Alarm	FanAlarm	fan, sensor, alarm	BI		Normal/Alarm	COV
Occupied Mode	Occ	occupied, cmd	BV	x	Occ/UnOcc	COV
Economize Mode	EconomizeMode	economize, run, cmd	BO		Active/Inactive	COV

Description	Point Name	Tags	Point Type	Override	UoM	History
Economize Cool Setpoint	EconomizeSp	economize, sp	AV	x	F	COV
Economize Enable Setpoint	EconomizeSp	economize, sp	AV		F	COV
RA CO2	RaCO2	return, co2, sensor	AI		ppm	Int,10min
RA Max CO2 Setpoint	RaMaxCO2Sp	return, co2, sp	AV	x	ppm	COV
RA Min CO2 Setpoint	RaMinCO2Sp	return, co2, sp	AV	x	ppm	COV

Electric Meter Point List Requirements

Point Desc	Point Name	Type	UoM	Override	History	Tags
Energy Total (accumulated)	Energy	AI	kWh	N	Int, 10min	elec, total, energy, sensor
Power Total	Power	AI	kW	N	Int, 10min	elec, total, power, active, sensor

Exhaust Fan Point List Requirements

Point Desc	Point Name	Type	UoM	Override	History	Tags
Exhaust Fan Command	EFCmd	BO	On/Off	Y	COV	fan, run, cmd
Exhaust Fan Status	EFSts	BI	On/Off	N	COV	fan, run, sensor
Zone Temperature	ZoneTemp	AI	F	N	Int, 15min	zone, temp, sensor
Zone Temperature Setpoint	ZoneTempSp	AO	F	Y	COV	zone, temp, sp
Exhaust Damper Command	EFDmprCmd	BO	Open/Closed	Y	COV	damper, cmd
Exhaust Damper Status	EFDmprSts	BI	Open/Closed	N	COV	damper, sensor
Alarm	EFAAlarm	BI	Normal/Alarm	N	COV	fan, alarm

Hotwater System Point List Requirements

Description	Point Name	Tags	Point Type	UoM	History	Override
Hot Water Supply Temp	HWSTemp	hot, water, leaving, temp, sensor	AI	F	Int, 10min	
Hot Water Supply Temp Setpoint	HWSTempSp	hot, water, leaving, temp, sp	AV	F	COV	x
Hot Water Return Temp	HWRTemp	hot, water, entering, temp, sensor	AI	F	Int, 10min	

Description	Point Name	Tags	Point Type	UoM	History	Override
Hot Water Alarm Setpoint	HWRAlarmSp	hot, water, leaving, temp, sp	AV	F	COV	x
Building Loop Pump Command	HWPCmd	hot, water, pump, run, cmd	BO	On/Off	COV	x
Building Loop Pump Status	HWPSs	hot, water, pump, run, sensor	BI	On/Off	COV	
Outside Air Temperature	OaTemp	outside, air, temp, sensor	AI	F	Int, 10 min	
Outside Air Temperature Setpoint	OaTempSp	outside, air, temp, sensor, sp	AV	F	COV	
Night Low Limit Setpoint	NightLowLimitSp	outside, air, temp, sensor, sp	AV	F	COV	

VAV Point List Requirements

Description	Point Name	Tags	Point Type	UoM	History	Override
Occupied Mode	Occ	occupied, cmd	BV	Occ/UnOcc	COV	x
Discharge Air Temperature	DaTemp	discharge, air, temp, sensor	AI	F	Int, 10min	
Reheat Valve	HWVlvPos	hot, water, valve, cmd	AO	%	Int, 10min	x
Cooling Max Flow	CoolMaxFlow	cooling, max, flow, sp	AV	cfm	Int, 15min	x
Cooling Min Flow	CoolingMin Flow	cooling, min, flow, sp	AV	cfm	Int, 15min	x
Heating Max Flow	HeatMaxFlow	heating, max, flow, sp	AV	cfm	Int, 15min	x
Heating Min Flow	HeatMinFlow	heating, min, flow, sp	AV	cfm	Int, 15min	x
AHU Supply Temp	AhuSupTemp	discharge, air, temp, sensor	AI	F	Int, 15Min	
Actual Discharge Air Flow	DaFlow	discharge, air, flow, sensor	AI	cfm	Int, 10min	
Discharge Air Flow Setpoint	DaFlowSp	discharge, air, flow, effective, sp	AV	cfm	COV	X
Zone Temperature	ZoneTemp	zone, air, temp, sensor	AI	F	Int, 10min	
Zone Temperature Setpoint	ZoneTempSp	zone, air, temp, effective, sp	AV	F	COV	x
Occupied Cooling Setpoint	OccCoolSp	occ, cooling, zone, air, temp, sp,	AV	F	COV	x
Occupied Heating Setpoint	OccHeatSp	occ, heating, zone, air, temp, sp	AV	F	COV	x

Description	Point Name	Tags	Point Type	UoM	History	Override
Unoccupied Cooling Setpoint	UnoccCoolSp	unocc, cooling, zone, air, temp, sp	AV	F	COV	x
Unoccupied Heating Setpoint	UnoccHeatSp	unocc, heating, zone, air, temp, sp	AV	F	COV	x
Zone Humidity	ZoneHumidity	zone, air, temp, sensor	AV	%RH	Int, 10min	
Damper Position	DmprPos	air, damper, cmd	AO	%	Int, 10min	

Constant Volume Air Handler Unit Point Requirements

Description	Point Name	Tags	Point Type	UoM	History	Override
Status of supply fan	SFanSts	fan, run, sensor	BI	On/Off	COV	
Command to start/stop supply fan	SFanCmd	fan, run, cmd	BO	On/Off	COV	x
Measured supply air temperature	DaTemp	discharge, air, temp, sensor	AI	F	Int, 10min	
Measured return air temperature	RaTemp	mixed, air, temp, sensor	AI	F	Int, 10min	
Measured mixed air temperature	MaTemp	mixed, air, temp, sensor	AI	F	Int, 10min	
Cooling Valve Control	CoolValveCmd	cool, water, valve, cmd	AO	%	Int, 10min	x
CoolWater Valve Position	CoolWtrPos	cool, water, valve, cmd	AI	%	Int, 10min	X
Heating Valve Control	HeatValveCmd	hot, water, valve, cmd	AO	%	Int, 10min	x
Hot Water Valve Position	HWVlvPos	hot, water, valve, cmd	AI	%	Int, 10min	
Cooling Water Setpoint	CwsTempSp	cool,water,sp	AV	F	COV	x
Heating Water Setpoint	HwsTempSp	hot,water, sp	AV	F	COV	x
Cooling Stage X (1,2, ...) Command	CoolStage(X)	runCmd, compressor, equip	BO	On/Off	COV	x
Cooling Stage X (1,2,...) Status	CoolSts(X)	runStatus, stage, compressor, equip	BI	On/Off	COV	
Freeze protection status	FreezeStat	freeze, sensor, alarm, ahu	BI	Normal/Alarm	COV	
Return or space CO2 reading	ReturnCO2	return, co2, sensor	AI	ppm	Int, 10min	
Outdoor air damper Position	OaDmprPos	outside, air, damper, cmd	AO	%	Int, 10min	x
Enables economizer operation	EconEnable	economize, run, cmd	BO	Enable/Disable	COV	X

Description	Point Name	Tags	Point Type	UoM	History	Override
Economize Cool Setpoint	EconomizeSp	economize, sp	Num	F	COV	x
Economize Enable Setpoint	EconomizeSp	economize, sp	Num	F	COV	x
Occupied Mode	Occ	occupied, cmd	BV	Occ/UnOcc	COV	x
DA Temp Setpoint	DaTempSp	discharge, air, temp, sp	AV	F	COV	x
Fire Alarm	FireAlarm	fire, alarm	BI	Normal/Alarm	COV	
RA Max CO2 Setpoint	RaMaxCO2Sp	return, co2, sp	AV	ppm	COV	x
RA Min CO2 Setpoint	RaMinCO2Sp	return, co2, sp	av	ppm	COV	x
Zone Temperature Setpoint	ZoneTempSp	zone, air, temp, effective, sp	AV	F	COV	x
Occupied Cooling Setpoint	OccCoolSp	occ, cooling, zone, air, temp, sp,	AV	F	COV	x
Occupied Heating Setpoint	OccHeatSp	occ, heating, zone, air, temp, sp	AV	F	COV	x
Unoccupied Cooling Setpoint	UnoccCoolSp	unocc, cooling, zone, air, temp, sp	AV	F	COV	x
Unoccupied Heating Setpoint	UnoccHeatSp	unocc, heating, zone, air, temp, sp	AV	F	COV	x
Zone Humidity	ZoneHumidity	zone, air, temp, sensor	AV	%RH	Int, 10min	

J. GRAPHICS

The City of Raleigh shall approve a draft version of the graphics for use at Green Road Community Center. The contractor is responsible for scheduling a graphic review meeting with the City of Raleigh Building Automation team, tracking feedback, and implementing changes to the graphics based on the feedback.

K. Room Network Temperature and Humidity Sensor

The sensor shall feature embedded BACnet communications for efficient monitoring and control. It shall accurately measure room temperature and relative humidity (RH) and include up/down setpoint control, a local override function, and the ability to be overridden via BACnet or a JACE controller (Override Status: Via BACnet to set limits on user up/down setpoint control). The device connects to an RS-485 MS/TP network, providing a single-point solution. Features include a backlit LCD and a user menu with password control.

1. POWER SUPPLY 20-28 Vac/dc (non-isolated half-wave rectified)
2. CONSUMPTION 35 mA max @ 24 Vdc
3. PROTECTION CIRCUITRY Reverse voltage protected, overvoltage protected
4. OPERATING CONDITIONS 0 to 55°C (32 to 131°F), 0-95 %RH non-condensing
5. STORAGE CONDITIONS -20 to 70°C (-4 to 158°F)
6. SENSOR COVERAGE AREA 100 m2 (1000 ft2) typical

7. ENCLOSURE White ABS, IP30 (NEMA 1)
8. ENCLOSURE DIMENSIONS 84mm W x 119mm H x 29mm D (3.3" x 4.7" x 1.15")
WIRING CONNECTIONS Screw terminal block (14 to 22 AWG)
9. COMMUNICATION INTERFACE Hardware: 2 wire RS-485 Software: Native BACnet® or Modbus MS/TP protocol, menu selectable Baud Rate: Locally set from 300 to 76800
MAC Address Range: BACnet® - 0-127 Modbus - 1-255 (Factory default is 2) (63 devices max on one daisy chain)
10. LCD DISPLAY Resolution: 0.5 or 1°C/°F selectable, 1 %RH Size: 38.1mm W x 16.5mm H (1.5" x 0.65"), 3 digit Backlight: Auto-dimming, enable/disable via jumper Viewed Values: Temperature Only, RH Only or alternating Temperature/RH (RH requires optional RH signal)
11. TEMPERATURE SIGNAL Sensing Element: 10,000Ω thermistor, ±0.2°C (0.4°F) Range: 0 to 50°C (32 to 122°F)
12. SETPOINT CONTROL User Interface: Front panel Up/Down buttons available via BACnet® or Modbus Setpoint Mode: Temperature (°C/°F) or RH, menu selectable (factory default is Temp & °C) Adjustable Setpoint Range: 10 to 50°C, 50 to 122°F or 10 to 85 %RH, menu selectable (factory default is 18 to 24°C) Minimum Span: 4°C/°F for 10 %RH Temperature Setpoint Resolution: 0.5° or 1°, menu selectable (factory default is 1°)
13. OVERRIDE SWITCH User Interface: Front panel button available via BACnet® or Modbus Override Status: Via BACnet® or Modbus "OCC" segment lights on LCD

Preferred Room Network Temperature and Humidity Sensor:

The preferred sensor shall be the Greystone NTRC with the following specifications:

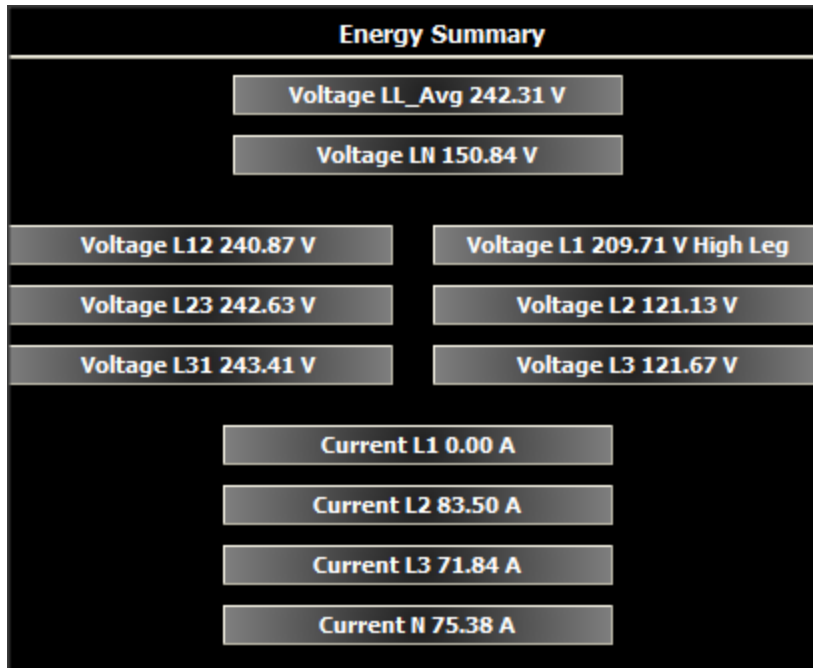
- **LCD Display:** L (Viewable)
- **Configuration:** RH (Temperature and Humidity)
- **Options:** P (Setpoint Adjustment), S (Override Switch)

Alternative room network temperature and humidity sensors may be used but must be approved by the City of Raleigh Building Automation Specialist.

L. Site Power Consumption

The Control contractor will provide similar power data collection currently implemented at Green Road Community Center.

1. BACnet/IP will be the required communication protocol for all power and gas metering devices within the project.



M. Schedule

The contractor shall implement one Niagara schedule to occupy and un-occupy the site.

N. Alarm Extensions

The contractor shall maintain the existing JACE-level alarm extensions and alarm configurations for all replacement points. Replacement points shall retain the same alarm functionality, settings, and alarm processing within the JACE as the points being replaced.

Old Point Extensions

Path	Name
/Drivers/BacnetNetwork/AHU\$2d2\$20GPC\$2d1/points/AHU2\$20FAN\$20PROOF\$2dalarmValue	CriticalAlarms_AHU2_SF
/Drivers/BacnetNetwork/AHU\$2d2\$20GPC\$2d1/points/AHU2\$20FAN\$20PROOF\$2dalarmValue	StatusAlarm
/Drivers/BacnetNetwork/AHU\$2d2\$20GPC\$2d1/points/AHU2\$20HEAT1\$20PROOF\$2dalarmValue	StatusAlarm
/Drivers/BacnetNetwork/AHU\$2d2\$20GPC\$2d1/points/AHU2\$20HEAT1\$20PROOF\$2dalarmValue	CriticalAlarms_AHu2_Heat1Proof
/Drivers/BacnetNetwork/AHU\$2d2\$20GPC\$2d1/points/AHU2_SpcTemp	New\$20Gym\$20temp\$20alarm
/Drivers/BacnetNetwork/AHU\$2d2\$20GPC\$2d1/points/AHU2_SpcTemp	BusinessHours_NewGym_SpcTmp
/Drivers/BacnetNetwork/AHU\$2d2\$20GPC\$2d1/points/Heat\$20Alarm	CriticalAlarms_Ahu2_HeatAlarm
/Drivers/BacnetNetwork/AHU\$2d2\$20GPC\$2d1/points/Heat\$20Alarm	alarm

/Drivers/BacnetNetwork/AHU\$2d2\$20GPC\$2d1/points/Run\$20Hours	BusinessHours_AHU2_RunHrs
/Drivers/BacnetNetwork/AHU\$2d2\$20GPC\$2d1/points/Run\$20Hours	AH2\$20Run\$20Hours\$20Alarm
/Drivers/BacnetNetwork/AHU\$2d3\$20GPC\$2d1/points/AHU3\$20FAN\$20PROOF\$2dalarmValue	CriticalAlarm_AHU3_SF_Proof
/Drivers/BacnetNetwork/AHU\$2d3\$20GPC\$2d1/points/AHU3\$20FAN\$20PROOF\$2dalarmValue	StatusAlarm
/Drivers/BacnetNetwork/AHU\$2d3\$20GPC\$2d1/points/AHU3\$20HEAT1\$20PROOF\$2dalarmValue	StatusAlarm
/Drivers/BacnetNetwork/AHU\$2d3\$20GPC\$2d1/points/AHU3\$20HEAT1\$20PROOF\$2dalarmValue	CriticalAlarm_AHU3_Heat1Proof
/Drivers/BacnetNetwork/AHU\$2d3\$20GPC\$2d1/points/AHU3_GYM_FAN_ALARM	BooleanChangeOfStateAlarmExt
/Drivers/BacnetNetwork/AHU\$2d3\$20GPC\$2d1/points/AHU3_SpcTemp	TEMP\$20ALARM
/Drivers/BacnetNetwork/AHU\$2d3\$20GPC\$2d1/points/AHU3_SpcTemp	BusinessHours_AHU3_SpcTmp
/Drivers/BacnetNetwork/AHU\$2d3\$20GPC\$2d1/points/Heat\$20Alarm	CriticalAlarms_AHU3_Heat
/Drivers/BacnetNetwork/AHU\$2d3\$20GPC\$2d1/points/Heat\$20Alarm	alarm
/Drivers/BacnetNetwork/AHU\$2d3\$20GPC\$2d1/points/Run\$20Hours	BusinessHours_AHU3_Runhrs
/Drivers/BacnetNetwork/AHU\$2d3\$20GPC\$2d1/points/Run\$20Hours	AH3\$20Run\$20Hours\$20Alarm
/Drivers/BacnetNetwork/Gym_AHU/points/Alarms/Alarm_Relay_Output_Status	BooleanChangeOfStateAlarmExt
/Drivers/BacnetNetwork/HEAT\$20PUMP\$201/points/HP1_SpcTmp	BusinessHours_HeatPump1
/Drivers/BacnetNetwork/HEAT\$20PUMP\$201/points/HP1_SpcTmp	TEMP\$20ALARM
/Drivers/BacnetNetwork/RTU\$2d4A_OLD/points/Fan\$20Event\$20State	CriticalAlarms_RTU4A_Fan
/Drivers/BacnetNetwork/RTU\$2d4A_OLD/points/Fan\$20Event\$20State	StatusAlarm
/Drivers/BacnetNetwork/RTU\$2d4A_OLD/points/RTU4A_SpcTmp	BusinessHours_RTU4A_SpcTmp
/Drivers/BacnetNetwork/RTU\$2d4A_OLD/points/RTU4A_SpcTmp	TEMP\$20ALARM
/Drivers/BacnetNetwork/RTU\$2d4B/points/Fan\$20Event\$20State	StatusAlarm
/Drivers/BacnetNetwork/RTU\$2d4B/points/Fan\$20Event\$20State	CriticalAlarm_RTU4B_Fan
/Drivers/BacnetNetwork/RTU\$2d4B/points/RTU4B_SpcTmp	BusinessHours_RTU4B_SpcTmp
/Drivers/BacnetNetwork/RTU\$2d4B/points/RTU4B_SpcTmp	TEMP\$20ALARM
/Drivers/BacnetNetwork/RTU\$2d5/points/Fan\$20event\$20state	StatusAlarm
/Drivers/BacnetNetwork/RTU\$2d5/points/Fan\$20event\$20state	CriticalAlarm_RTU5_Fan
/Drivers/BacnetNetwork/RTU\$2d5/points/RTU5_SpcTmp	TEMP\$20ALARM
/Drivers/BacnetNetwork/RTU\$2d5/points/RTU5_SpcTmp	BusinessHours_RTU5_SpcTmp
/Drivers/BacnetNetwork/RTU\$2dLOBBY/points/BooleanPoint	CriticalAlarms_RTULobby_Fan
/Drivers/BacnetNetwork/RTU\$2dLOBBY/points/BooleanPoint	StatusAlarm

/Drivers/BacnetNetwork/RTU\$2dLOBBY/points/RTU\$2dLOBBY_SpcT mp	BusinessHours_RTU_Lobby_SpcTmp
/Drivers/BacnetNetwork/RTU\$2dLOBBY/points/RTU\$2dLOBBY_SpcT mp	TEMP\$20ALARM
/Drivers/BacnetNetwork/RTU_4A/points/ZoneTemp	TEMP\$20ALARM
/Services/AlarmService/TestAlarm	BooleanChangeOfStateAlarmExt

O. Enterprise Developer

1. Dashboards

- Power
 - a. Total KWH Per Year; keep up to three years of data.
 - b. Total KWH use today.
 - c. Total KWH used yesterday.
 - d. Total KWH use per month.
- Average Space Temperature compared to Cooling and Heating Setpoints.
 - a. Monthly Average Temperature per AHU, monthly.
 - b. Monthly Average Temperature Per VAV, monthly.
 - c. Keep up three years of data.
- Add Green Road Community Center to the City_of_Raleigh Periscope Map.

2. Point Integration – Old Name to New Name

The Enterprise Developer will be responsible for integrating the points from the Green Road Community Center Niagara station with the City_of_Raleigh Supervisor station. The old point names must be migrated to the new point names listed in Section 1.31(F). The Enterprise Developer shall ensure that the same path on the Supervisor Station is reused for all points below.

Old Point Names

Path	Name
/Drivers/NiagaraNetwork/GreenRdRec/points/ New Gym Space TEMP2	New Gym Space TEMP2
/Drivers/NiagaraNetwork/GreenRdRec/points/ NEW GYM SUPPLY TEMP2	NEW GYM SUPPLY TEMP2
/Drivers/NiagaraNetwork/GreenRdRec/points/ NEW GYM SPACE TEMP3	NEW GYM SPACE TEMP3
/Drivers/NiagaraNetwork/GreenRdRec/points/ NEW GYM SUPPLY TEMP3	NEW GYM SUPPLY TEMP3
/Drivers/NiagaraNetwork/GreenRdRec/points/ NEW LOBBY SPACE TEMP	NEW LOBBY SPACE TEMP
/Drivers/NiagaraNetwork/GreenRdRec/points/ NEW LOBBY SUPPLY TEMP	NEW LOBBY SUPPLY TEMP
/Drivers/NiagaraNetwork/GreenRdRec/points/ DANCE 4A SPACE TEMP	DANCE 4A SPACE TEMP
/Drivers/NiagaraNetwork/GreenRdRec/points/ DANCE 4A SUPPLY TEMP	DANCE 4A SUPPLY TEMP
/Drivers/NiagaraNetwork/GreenRdRec/points/ DANCE 4B SPACE TEMP	DANCE 4B SPACE TEMP
/Drivers/NiagaraNetwork/GreenRdRec/points/ DANCE 4B SUPPLY TEMP	DANCE 4B SUPPLY TEMP
/Drivers/NiagaraNetwork/GreenRdRec/points/ CLASS ROOM 5 SPACE TEMP	CLASS ROOM 5 SPACE TEMP
/Drivers/NiagaraNetwork/GreenRdRec/points/ CLASS ROOM 5 SUPPLY TEMP	CLASS ROOM 5 SUPPLY TEMP
/Drivers/NiagaraNetwork/GreenRdRec/points/ LOBBY SPACE TEMP	LOBBY SPACE TEMP
/Drivers/NiagaraNetwork/GreenRdRec/points/ LOBBY SUPPLY TEMP	LOBBY SUPPLY TEMP
/Drivers/NiagaraNetwork/GreenRdRec/points/ OLD GYM SPACE TEMP	OLD GYM SPACE TEMP

/Drivers/NiagaraNetwork/GreenRdRec/points/ OLD GYM SUPPLY TEMP	OLD GYM SUPPLY TEMP
/Drivers/NiagaraNetwork/GreenRdRec/points/ OAT	OAT
/Drivers/NiagaraNetwork/GreenRdRec/points/ GreenRd AHU2 OAD	GreenRd AHU2 OAD
/Drivers/NiagaraNetwork/GreenRdRec/points/ GreenRd AHU3 OAD	GreenRd AHU3 OAD
/Drivers/NiagaraNetwork/GreenRdRec/points/ GreenRd RTU4A OAD	GreenRd RTU4A OAD
/Drivers/NiagaraNetwork/GreenRdRec/points/ GreenRd RTU4B OAD	GreenRd RTU4B OAD
/Drivers/NiagaraNetwork/GreenRdRec/points/ GreenRd RTU5 OAD	GreenRd RTU5 OAD
/Drivers/NiagaraNetwork/GreenRdRec/points/ GreenRd RTULobby OAD	GreenRd RTULobby OAD

Old Point Names Zone Temperature

Path	Name
/Drivers/NiagaraNetwork/GreenRdRec/points/ZoneTemps/ AHU2_SpcTemp	AHU2_SpcTemp
/Drivers/NiagaraNetwork/GreenRdRec/points/ZoneTemps/ AHU2_HeatSpt	AHU2_HeatSpt
/Drivers/NiagaraNetwork/GreenRdRec/points/ZoneTemps/ AHU2_CoolSpt	AHU2_CoolSpt
/Drivers/NiagaraNetwork/GreenRdRec/points/ZoneTemps/ AHU3_SpcTemp	AHU3_SpcTemp
/Drivers/NiagaraNetwork/GreenRdRec/points/ZoneTemps/ AHU3_HeatSpt	AHU3_HeatSpt
/Drivers/NiagaraNetwork/GreenRdRec/points/ZoneTemps/ AHU3_CoolSpt	AHU3_CoolSpt
/Drivers/NiagaraNetwork/GreenRdRec/points/ZoneTemps/ HP1_SpcTmp	HP1_SpcTmp
/Drivers/NiagaraNetwork/GreenRdRec/points/ZoneTemps/ HP1_EffSPt	HP1_EffSPt
/Drivers/NiagaraNetwork/GreenRdRec/points/ZoneTemps/ RTU4A_SpcTmp	RTU4A_SpcTmp
/Drivers/NiagaraNetwork/GreenRdRec/points/ZoneTemps/ RTU4A_EffSpt	RTU4A_EffSpt
/Drivers/NiagaraNetwork/GreenRdRec/points/ZoneTemps/ RTU4B_SpcTmp	RTU4B_SpcTmp
/Drivers/NiagaraNetwork/GreenRdRec/points/ZoneTemps/ RTU4B_EffSpt	RTU4B_EffSpt
/Drivers/NiagaraNetwork/GreenRdRec/points/ZoneTemps/ RTU5_SpcTmp	RTU5_SpcTmp
/Drivers/NiagaraNetwork/GreenRdRec/points/ZoneTemps/ RTU5_EffSpt	RTU5_EffSpt
/Drivers/NiagaraNetwork/GreenRdRec/points/ZoneTemps/ RTU-LOBBY_SpcTmp	RTU-LOBBY_SpcTmp
/Drivers/NiagaraNetwork/GreenRdRec/points/ZoneTemps/ RTU-LOBBY_EffSpt	RTU-LOBBY_EffSpt
/Drivers/NiagaraNetwork/GreenRdRec/points/ZoneTemps/ RTU-GYM_HeatSpt	RTU-GYM_HeatSpt
/Drivers/NiagaraNetwork/GreenRdRec/points/ZoneTemps/ RTU-GYM_CoolSpt	RTU-GYM_CoolSpt
/Drivers/NiagaraNetwork/GreenRdRec/points/ZoneTemps/ RTU-GYM_SpcTmp	RTU-GYM_SpcTmp

3. Histories Stored On City_of_Raleigh Supervisor Station

The Enterprise Developer shall configure the City_of_Raleigh Supervisor station to upload all history data from the Niagara station for Green Road CC. The execution time to upload history data shall be 2:00 AM (Sun Mon Tue Wed Thu Fri Sat).

Status	{ok}
State	Idle
Enabled	☒ true
Execution Time	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
Last Attempt	20-Feb-2025 02:00 AM EST
Last Success	20-Feb-2025 02:00 AM EST
Last Failure	03-Jan-2025 12:57 PM EST
Fault Cause	
History Id	/ ^ / AHU1 COOL 1 K2,CV
On Demand Poll Enabled	☒ true
On Demand Poll Frequency	Normal
Config Overrides	Component

Existing History Data On Supervisor Station for Green Road CC

Name	History Id	Execution Time
4A Dance Rm Space Tmp Trend	^4A Dance Rm Space Tmp Trend	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
4B Dance Rm Space Tmp Trend	^4B Dance Rm Space Tmp Trend	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
AHU2 Fan proof	^AHU2 Fan proof	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
AHU2 comp 2 proof	^AHU2 comp 2 proof	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
AHU3_COOLING STPT CALC	^AHU3_COOLING STPT CALC	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
COOLING STPT CALC	^COOLING STPT CALC	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
GYM TREND	^GYM TREND	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
HEAT PUMP 1 ZONE TEMP	^HEAT PUMP 1 ZONE TEMP	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
HEATING STPT CALC	^HEATING STPT CALC	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
HP1_EffSpt	^HP1_EffSpt	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
HWSPUMP1 TREND	^HWSPUMP1 TREND	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
KWH Hourly Multiply	^KWH Hourly Multiply	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
NEW AHU 2 OARH TREND	^NEW AHU 2 OARH TREND	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
NEW AHU 2 RAT TREND	^NEW AHU 2 RAT TREND	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}

NEW AHU 2 SAT TREND	^NEW AHU 2 SAT TREND	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
NEW AHU 2 SPC TEMP TREND	^NEW AHU 2 SPC TEMP TREND	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
NEW AHU 3 SAT TREND	^NEW AHU 3 SAT TREND	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
NEW AHU 3 SPC TEMP TREND	^NEW AHU 3 SPC TEMP TREND	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
OAT TREND	^OAT TREND	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
Old Gym Space Tmp Trend	^Old Gym Space Tmp Trend	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
RTU 5 Class Rm Space Tmp Trend	^RTU 5 Class Rm Space Tmp Trend	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
RTU 7 Lobby Space Tmp Trend	^RTU 7 Lobby Space Tmp Trend	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
RTU-GYM_CoolSpt	^RTU-GYM_CoolSpt	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
RTU-GYM_HeatSpt	^RTU-GYM_HeatSpt	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
RTU-LOBBY_EffSpt	^RTU-LOBBY_EffSpt	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
RTU4A_EffSpt	^RTU4A_EffSpt	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
RTU4B_EffSpt	^RTU4B_EffSpt	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
RTU5_EffSpt	^RTU5_EffSpt	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
RTU_4A_Cool1	^RTU_4A_Cool1	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
RTU_4A_ExhaustFanStatus	^RTU_4A_ExhaustFanStatus	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
RTU_4A_FanStatus	^RTU_4A_FanStatus	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
RTU_4A_Heat1	^RTU_4A_Heat1	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
RTU_4A_OccCoolSpt	^RTU_4A_OccCoolSpt	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
RTU_4A_OccHeatSpt	^RTU_4A_OccHeatSpt	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
RTU_4A_ZoneEffSpt	^RTU_4A_ZoneEffSpt	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}
STATIC PR. TREND	^STATIC PR. TREND	2:00 AM {Sun Mon Tue Wed Thu Fri Sat}

4. Provisioning Station Backup

The Enterprise Developer will be responsible for configuring the Niagara station provisioning backup, allowing the Supervisor station to schedule regular backups of the Green Road CC Niagara station.

5. FOXS Niagara Network

The Enterprise Developer will be responsible for configuring the Niagara Framework to enable secure (**FOXs**) real-time data exchange between the **City_of_Raleigh** Supervisor station and the **Green Road CC** Niagara station.

P. Air Handler Freeze Protection

1. The control program for each air handler shall implement a freeze protection safety feature to prevent coil freezing.
2. The contractor shall implement a freeze stat alarm (BI) – freeze protection alarm status for all air handler units.

Q. JACE Time Server Integration

The JACE controller shall be configured to synchronize with the site's designated time server. Verify proper time synchronization and confirm that system schedules, trends, alarms, and time-based sequences operate using the correct system time. Work with COR IFS team member to get IP address for time server used.

1.32 ELECTRICAL MISCELLANEOUS

- A. Panels: All enclosures for DDC controllers and devices shall be fabricated in accordance with UL Standards from code gauge steel. Enclosures shall be provided with a continuous hinge on the door and a flush latching mechanism. Enclosures shall be shop painted with standard grade enamel coating. Back panels shall be furnished when required to facilitate installation of boards or accessories. All enclosures installed outdoors shall be constructed to NEMA 3R standards. All controllers shall be installed within an approved enclosure unless the controller will be installed within the control cabinet section of the equipment that it is intended to control. Enclosures shall facilitate the mounting of gauges, switches, pilot lights, and the like, on the face panel when required. Control devices that are mounted on the face of the panel shall be identified with engraved nameplates. Panels shall be Hoffman A1 series or approved equal.
- B. Power Transformers: Step-down power transformers shall be provided for all DDC controllers and associated accessory devices as required. Transformers shall be sized and selected to accommodate all connected accessory items. Transformers shall be UL Listed Class 2 type with 120 VAC primary, 24 VAC secondary. Transformers shall be Functional Devices TR series or approved equal.
- C. Relays: Miscellaneous control relays shall be provided as required to energize or control equipment and devices within the control system. Relays shall be located as close as practical to the controlled device (motor, motor starter, etc.). Where approved by NEC, relays may be installed within starters and equipment control panels where space is available. Relays installed outside of the controlled device shall be provided with a NEMA enclosure suitable for the location where installed. Relays shall be Functional Devices RIB series or approved equal.

1.33 ELECTRICAL WIRING

- A. Wiring: All wiring devices and accessories shall comply with the requirements of Division 26 and the NEC. All wiring shall be installed in a neat and professional manner. Control wiring shall not be installed in power circuit conduits or raceways unless specifically approved for that purpose. All wiring, except plenum wiring (where allowed), shall be run in electrical conduits. Plenum cable will be allowed in concealed locations where accessible. All cable must be installed with 90° angles and strapped according the NEC.
- B. Provide all interlock and control wiring. Provide wiring as required by functions as specified and as recommended by equipment and device manufacturers to achieve the specified control functions.

- C. Low voltage conductors shall be stranded bare or tinned-copper with premium grade polymer alloy insulation. For shielded cable, furnish multi-conductor of overall polyester supported aluminum foil with stranded tinned copper drain wire to facilitate grounding. Coaxial shield shall be copper braided type. Provide shielded cable where recommended by the equipment or device manufacturer, grounded in strict accordance with the manufacture's recommendations.
- D. Magnetic starters and disconnect switches shall not be used as junction boxes. Provide auxiliary junction boxes as required. Terminations for Fire Alarm Control Panel (FACP) interface shall be accomplished by the Electrical Contractor or his designated subcontractor.
- E. FMCS Systems Integrator shall provide power for all control devices and components from the closest available power source or as indicated on the power Drawings. When acceptable to the equipment manufacturer, low voltage power may be obtained from the internal equipment power source or transformer. Electrical Power for Systems Integrator's use has been provided at j-boxes located on plans.
- F. Magnetic starters shall be furnished and installed by the Electrical Contractor.
- G. Disconnects shall be furnished and installed by the Electrical Contractor.

PART 3 - EXECUTION

1.34 INSTALLATION

- A. All work described in this section shall be performed by system integrators or contractors that have a successful history in the design and installation of integrated control systems. The installing office shall have a minimum of five years of integration experience and shall provide documentation in the submittal package verifying the company's experience.
- B. Install system and materials in accordance with manufacturer's instructions, and as detailed on the project drawing set.
- C. Drawings of FMCS network are diagrammatic only and any apparatus not shown but required to make the system operative to the complete satisfaction of the Architect shall be furnished and installed without additional cost.
- D. Line and low voltage electrical connections to control equipment shown specified or shown on the control diagrams shall be furnished and installed by the FMCS Systems Integrator in accordance with the specifications in Divisions 23 and 26.

1.35 WIRING

- A. All electrical control wiring and power wiring to the NAC, computers and network components shall be the responsibility of the FMCS contractor.
- B. All wiring shall be in accordance with the National Electrical Code and any applicable local codes. All FMCS wiring shall be installed in the conduit unless otherwise allowed by the National Electrical Code or applicable local codes. Where FMCS plenum rated cable wiring is

allowed it shall be run parallel to or at right angles to the structure, properly supported and installed in a neat and workmanlike manner.

1.36 WARRANTY

- A. Equipment, materials and workmanship incorporated into the work shall be warranted for a period of one year from the time of system acceptance.
- B. Within this period, upon notice by the Owner, any defects in the work provided under this section due to faulty materials, methods of installation or workmanship shall be promptly (within 48 hours after receipt of notice) repaired or replaced by the FMCS contractor at no expense to the Owner.

1.37 WARRANTY ACCESS

- A. Pending owner pre-approval, the Owner shall grant to the FMCS contractor, reasonable access to the FMCS during the warranty period. The owner shall allow the contractor to access the FMCS from a remote location for the purpose of diagnostics and troubleshooting, via the Internet, during the warranty period.

1.38 SOFTWARE LICENSE

- A. The Owner shall be the named license holder of all software associated with any and all incremental work on the project(s). The owner requires that all Vykon-based software and hardware on this project be the current version of N4.
- B. The owner, or his appointed agent, shall receive ownership of all job specific software configuration documentation, data files, and application-level software developed for the project. This shall include all custom, job specific software code and documentation for all configuration and programming that is generated for a given project and /or configured for use within Niagara4 based controllers and/or servers and any related LAN / WAN / Intranet and Internet connected routers and devices. Any and all required Ids and passwords for access to any component or software program shall be provided to the owner.
- C. Provide documentation verifying Niagara licensing information, including confirmation that the Customer-Owned Repository (COR) is owned and controlled by the Owner. All required licenses, certificates, and software entitlements shall be properly transferred and assigned prior to project completion.

1.39 ACCEPTANCE TESTING

- A. Upon completion of the installation, the FMCS contractor shall load all system software and start-up the system. The FMCS contractor shall perform all necessary calibration, testing and de-bugging and perform all required operational checks to insure that the system is functioning in full accordance with these specifications. The FMCS contractor shall coordinate the checkout of the system such that other appropriate Divisions have a representative present during system checkout.

- B. The FMCS contractor shall perform tests to verify proper performance of components, routines, and points. Repeat tests until proper performance results. This testing shall include a point-by-point log to validate 100% of the input and output points of the DDC system operation.
- C. Upon completion of the performance tests described above, repeat these tests, point by point as described in the validation log above in presence of Owner's Representative, as required. Properly schedule these tests so testing is complete at a time directed by the Owner's Representative. Do not delay tests so as to prevent delay of occupancy permits or building occupancy.
- D. System Acceptance: Satisfactory completion is when the Division 230900 contractor has performed successfully all the required testing to show performance compliance with the requirements of the Contract Documents to the satisfaction of the Owner's Representative. System acceptance shall be contingent upon completion and review of all corrected deficiencies.

1.40 OPERATOR INSTRUCTION, TRAINING

- A. During system commissioning and at such time acceptable performance of the FMCS hardware and software has been established, the contractor shall provide on-site operator instruction to the owner's operating personnel. Operator instruction shall be done during normal working hours and shall be performed by a competent representative familiar with the system hardware, software and accessories.
- B. The FMCS contractor shall provide 24 hours of instruction to the owner's designated personnel on the operation of the FMCS and describe its intended use with respect to the programmed functions specified. Operator orientation of the FMCS shall include, but not be limited to; the overall operation program, equipment functions (both individually and as part of the total integrated system), commands, systems generation, advisories, and appropriate operator intervention required in responding to the System's operation.
- C. The training shall be in three sessions as follows:
 - 1. One day session (8 hours) after system is started up and at least one week before first acceptance test. Manual shall have been submitted at least two weeks prior to training so that the owners' personnel can start to familiarize themselves with the system before classroom instruction begins.
 - 2. One day follow up session (8 hours) after system has been operational for at least 1 month.
 - 3. Warranty Follow Up: Two days (4 hours each) to be scheduled at the request of the owner during the one-year warranty period. These sessions shall cover topics as requested by the owner such as; how to add additional points, create and gather data for trends, graphic screen generation or modification of control routines.

END OF SECTION 230900

SECTION 233100 - DUCTWORK

PART 1 - GENERAL

- 1.1 This Section includes ductwork, splitter dampers, balancing dampers, air deflection devices, etc. required for a complete system.
- 1.2 The Drawings are intended to indicate, with reasonable accuracy, the location of components and the general arrangement of the system. All offsets, bends fittings and other devices, not shown but required for the full operation of the system, shall be provided.
- 1.3 Refer to specification Section 230700 for duct insulation

PART 2 - PRODUCT

- 2.1 Low and Medium Pressure Ductwork.
 - A. Round and rectangular ductwork shall be of gauges and construction methods as indicated in the latest ASHRAE Guide and SMACNA Standard.
 - B. Splitter dampers, balancing dampers, turning vanes and air deflection devices shall be installed as shown on the plans and/or where required for the proper control of airflow.
 - C. All take-offs to diffusers shall be tapered type taps with factory damper and locking quadrant.
 - D. All take-offs to VAV Units shall be made with conical taps.
- 2.2 Flexible Ductwork
 - A. Ducts shall be insulated type with foil wrapper complying with NFPA Standard No. 90A and UL181.
 - B. All flexible ducts shall have a factory installed R8 fiberglass insulation with a seamless vinyl vapor barrier.
 - C. Length of flexible duct shall not exceed 10 feet.
 - D. Flexible duct shall be secured and sealed in place with mastic to hard duct collars at each end, with nylon tie-wraps on the wire enforced inner mylar skin, followed by the insulation layer and then the exterior vapor layer secured with another tie-wrap.
- 2.3 Exposed Ductwork
 - A. Exposed shall be round, 18 gauge spiral lock seam with paintable finish, double wall and internally insulated at the factory. Inner wall shall be perforated.
 - B. Duct shall be fastened using sheet metal screws only and no duct tape.

PART 3 - EXECUTION

- 3.1 Turning vanes shall be installed in square elbows for all ductwork.

- 3.2 Duct transitions, splitter dampers, and balancing dampers shall be constructed of gauges and materials as indicated in ASHRAE Guide and SMACNA Standards.
- 3.3 Hangers and supports for ductwork shall be of metal bands, angles and rods as indicated in ASHRAE Guide and SMACNA Standards. The minimum bandwidth shall be 1", 16 gauge, galvanized steel.
- 3.4 Where ductwork passes through floors and walls, the space around the ducts shall be sealed in an approved manner with mineral wool insulation, and/or proper fire seal material approved by the State or Local Inspector.
- 3.5 In exposed areas and mechanical rooms, ductwork openings shall be finished with a metal collar.
- 3.6 Ductwork shall be cross-braced and reinforced properly with galvanized steel angles as recommended by SMACNA Standards.
- 3.7 Where ductwork behind grilles or diffusers is visible, it shall be painted with two coats of flat black base fire retardant paint.
- 3.8 Duct connections to outside air louvers shall be pitched to drain outside and shall be soldered watertight.
- 3.9 Tape all low-pressure joints with Hardcast or approved equal for completely airtight system.
- 3.10 All medium pressure joints are to be sealed in accordance with SMACNA standards for ductwork 2" W.C. and greater. All ducts shall be air tight, rigid and free from vibration and noise.
- 3.11 Duct dimensions shown on the drawings are net inside dimensions.
- 3.12 Where ductwork is lined, as noted in Section 230700, the duct insulation thickness shall be added to the listed ductwork dimensions for final duct size.

END OF SECTION 233100

SECTION 233600 - AIR TERMINAL UNITS

PART 1 - GENERAL

- 1.1 The contractor shall provide air terminal units where shown on the plans.
- 1.2 Units shall be selected as scheduled on the plans.
- 1.3 Units shall be by Price, Nailor, Carrier, Trane, Metalaire, Reddi, or Titus.
- 1.4 Air terminal units shall be tested and certified in accordance with ARI Standard 880.

PART 2 - PRODUCT

2.1 VAV Terminal Unit

- A. The unit casing shall be of 24-gauge zinc coated steel, acoustically lined with $\frac{3}{4}$ " thick, 1-1/2 pound density fiberglass and shall comply with UL 181 and NFPA 90A.
- B. The reversible actuator shall be shaft mounted without linkage and shall be an integral part of the electronic controller.
- C. Units shall have pressure independent electronic control and shall be reset for air flow between zero and the maximum cataloged cfm.
- D. The damper shall be double thickness of 24-gauge steel, with shaft rotating in self-lubricating bearing.
- E. Dampers shall be designed such as to prevent air leakage in excess of 1% of the rated quantity at rated inlet static pressure.

2.2 Fan Terminal Unit

- A. The unit casing shall be of 20-gauge zinc coated steel, acoustically lined with 1" thick, 1 ½ pound density fiberglass and shall comply with UL 181 and NFPA 90A.
- B. Unit fans shall be forward curve centrifugal type with direct drive, three speed, permanently lubricated split capacitor type motors.
- C. Unit shall have factory-installed volume control dampers. Dampers shall be designed such as to prevent air leakage in excess of 1% of the rated air quantity at rated inlet static pressure.
- D. Terminal unit fan shall be controlled by means of an electronic speed controller with voltage limiting circuit and shall turn on whenever heating is required even though primary air system is turned off.
- E. Provide with 1" filter rack on the fan air inlet.

2.3 Hot Water Coils

- A. Coils are to be 1-row, constructed of aluminum fin and copper serpentine-type tubes with sweat connections tested at 250 psig.

PART 3 - EXECUTION

- 3.1 The units shall be installed where shown on the plans.
- 3.2 All units shall be installed according to manufacturer's recommendations.
- 3.3 Suspend units from spring and rubber vibration isolation hangers.
- 3.4 Provide flexible connections at ductwork and piping.
- 3.5 All electrical components shall be UL recognized and installed in accordance with the NEC.
- 3.6 Refer to drawings for controls.
- 3.7 Actuators shall be supplied by controls contractor and mounted at the factory. \
- 3.8 Label ceiling grid or access door where installed above ceiling.

END OF SECTION 233600

SECTION 238113 – OUTDOOR PACKAGED UNIT

A. GENERAL

1. Furnish and install where shown on the plan, a one-piece/cooling unit with capabilities as shown on the plans.
2. The unit shall be completely factory assembled, pre-charged, pre-wired, tested and ready to operate.
3. Unit shall be U.L. labeled.
4. Unit shall be Trane or approved equivalent by Carrier or York.

B. PRODUCT

1. Gas Heat/DX Cooling
 - a. Cabinet shall be single, enclosed, and weatherproof casing or galvanized steel bonderized and finished with baked enamel. Entire cooling section shall be fully insulated with fire retardant insulation to prevent sweating. A base pan drain connection shall be provided. Panels shall be easily removable for service access.
 - b. Compressor system shall consist of serviceable hermetic compressor. Compressor shall have service shut-off valves; suction pressure operated capacity control unloader, suitable vibration isolators and crankcase heater.
 - c. Condenser and evaporator coils shall have aluminum plate fins mechanically bonded to copper tubes.
 - d. Indoor air fans shall be forward curved, centrifugal type, belt driven. Outdoor fans shall be propeller type, direct driven. All motors shall have overload protection and suitable vibration isolators.
 - e. Cooling system shall be protected by fusible plug, high and low pressurestat, compressor motor overloads, anti-cycling timer device (5 minutes). Controls shall include low voltage control circuit transformer, compressor and fan motor safety controls with automatic reset, high and low pressure cutout switches and terminals for accessory electrical connections.
 - f. Gas heater assembly shall include 18-gauge aluminized steel heat exchanger, intermittent ignition, safety lockout, redundant gas valve, pressure switch, and mechanical combustion system.
 - g. Accessories shall be as indicated on the drawings.

C. EXECUTION

1. Units shall be located as shown on the plans.
2. Units shall be installed and connected in strict accordance with the manufacturer's installation instructions.
3. Controls shall be as indicated on the plans.

D. INSTALLATION, SERVICE, AND WARRANTY

1. The product manufacturer shall warrant his equipment for a period of one year to be free from defects in material and workmanship. Any part of the equipment that is found to be defective within the one year period shall be repaired and/or replaced by a well qualified factory designated repair station at no cost to the Owner.

2. The product manufacturer shall warrant the sealed refrigeration circuit for a period of five years. The sealed refrigeration circuit shall consist of the hermetic compressor assembly, evaporator coil, condenser coil, thermostatic expansion valve, and interconnecting tubing. All repairs under this warranty shall be made by a factory designated repair station at no cost to the Owner.
3. The warranty periods shall start from the day that the job is accepted by the Owner.

END OF SECTION 238113

SECTION 260000 – ELECTRICAL REQUIREMENTS

PART 1 - GENERAL

1.1 GENERAL NOTES:

- A. Drawings, all Contract Documents, and Division-1 Specifications sections, apply to work of this section.
- B. Where the term "Contractor" is used, it shall mean the Electrical Contractor.
- C. Contractors bidding on this section must possess a State of North Carolina Electrical Contractor's License. Each bidder shall show his license proposal.
- D. Reference shall be made to the Architectural, Structural, Plumbing and HVAC drawings and specifications for details of building construction and for coordination with other parts of construction.
- E. Contractor shall visit the job site before the submission of a bid and familiarize himself with existing conditions. Submission of a bid will be considered as evidence that the Contractor has visited the site and is familiar with existing conditions.

1.2 BIDDING:

The Electrical work shall be included under the General Contract.

1.3 SCOPE:

- A. The work under this section of the specifications consists of providing all labor, equipment, supplies, and materials, and performing all operations, including trenching, backfilling, compaction, cutting, channeling and chasing necessary for the installation of a complete wiring system in strict accordance with this section of the specifications and the applicable drawings and subject to the terms and conditions of the contract.
- B. The following gives a brief outline of the Electrical work which is further described in the body of the specifications and indicated on the accompanying drawings:
 - Electrical panel, and disconnect switches.
 - Feeder conduits, conductors, and fittings as shown and specified.
 - Lighting fixtures and lamps.
 - Telephone outlets.
 - All power and lighting branch conduit, conductors, outlets, boxes, receptacles, switches and circuit breakers as shown and specified.
 - Electrical power connections complete to all equipment requiring electrical connections.
 - Grounding
- C. Certified manufacturer's drawings and shop drawings shall be submitted for all equipment except the following for which lists of manufacturers and catalog numbers shall be submitted. Submit one hard copy with digital copies.
 - Conduit
 - Conduit fitting

Wiring devices

Wire

Boxes

- 1.4 APPLICABLE SPECIFICATIONS AND STANDARDS: The following standards shall be considered as minimum requirements for this project:

The National Electrical Code.

The National Electrical Safety Code.

The North Carolina State Building Code.

The Underwriter's Laboratories, Inc.

The National Electrical Manufacturers association.

The Insulated Power Cable Engineers Association.

NFPA 99 and 101 Life Safety Code for Health Care Occupancies

- 1.5 DRAWINGS AND SPECIFICATIONS:

- A. The drawings accompanying these specifications and subsequent drawings or detail will form a part of the contract for this work.
- B. The drawings are diagrammatic and indicate the extent and general arrangement of the outlets and equipment. These drawings shall be followed as closely as possible, but all measurements shall be verified at the job site. Do not scale drawings for location dimensions. Consult the architectural, structural, mechanical, plumbing and equipment drawings and details for this project for exact location of equipment and before starting the work.
- C. The drawings and specifications are supplementary, one to the other, and material and workmanship indicated, called for or implied by the one and not by the other shall be supplied and installed as if specifically called for by both.
- D. Omission of particular reference to any item necessary for a complete installation and proper operation thereof, shall not relieve the Contractor of the responsibility of furnishing same.
- E. Discrepancies shown on different drawings, between drawings and specifications or between plans and field conditions shall be promptly brought to the attention of the Engineer for a decision.

- 1.6 CODES, PERMITS AND INSPECTIONS:

- A. The entire installation shall comply with:

All laws and ordinances applying to electrical installations.

The requirements of the National Electrical Code (latest edition with applicable revisions) where such requirements do not conflict with the laws in effect.

The regulations of the power company furnishing the electrical service.

In the event of conflict between these codes, requirements, etc., the most stringent shall apply.

- B. The Contractor shall give all required notices, obtain necessary permits, and pay all required fees.
- C. After completion of the work, the Contractor shall furnish to the Owner a certificate of final inspection and approval from the electrical inspection department having jurisdiction.

1.7 SINGULAR:

In all cases where a device or piece of equipment is referred to herein or on the drawings in the singular number, it is intended that such reference shall apply to as many such devices as are required to complete the installation.

1.8 USE OF THE WORD "PROVIDE":

Herein, where the word "Provide" is written in these specifications, provide shall be understood to mean provide complete in place, that is, "Furnish and Install".

1.9 ELECTRIC SERVICE:

- A. Voltage: Refer to electrical plans.

1.10 EQUIPMENT AND MATERIALS:

- A. Catalog numbers and trade names in these specifications and noted on the drawings are intended to describe the material, devices or apparatus wanted. Similar materials, devices or apparatus of other manufacturers, if of equivalent quality, capacity and character, may be substituted on the written approval of the engineer. Proposed substitutions with descriptive data shall be submitted to the engineer at least ten days before the Bid Date. The submissions, if approved by the engineer, will be issued in an addendum before the Bid Opening. If the Contractor fails to comply with the provisions of this paragraph, he shall be required to furnish all materials and equipment as specified.
- B. All materials shall be new and shall bear the manufacturer's name, trade name and the UL label in every case where a standard has been established for the particular material. The equipment to be furnished shall be essentially the standard product of a manufacturer regularly engaged in the production of the required type of equipment and shall be the manufacturer's latest approved design.
- C. Equipment and materials shall be delivered to the site and stored in original containers, suitably sheltered from the elements but readily accessible for inspection until installed.
- D. Equipment and materials of the same general type shall be of the same make throughout the work to provide uniform appearance, operation and maintenance.
- E. Equipment shall be tightly covered and protected against dirt, water and chemical or mechanical injury and theft. Damage or defects developing before acceptance of the work shall be made good at the Contractor's expense.
- F. Dimensions: It shall be the responsibility of the Contractor to insure that items to be furnished fit the space available. He shall make necessary field measurements to ascertain space requirements, including those for connections, and shall furnish and install such sizes and shapes of equipment that the final installation shall suit the true intent and meaning of the drawings and specifications.

1.11 WORKMANSHIP:

- A. All work specified herein shall be executed in the best and most workmanlike manner by skilled mechanics. This Contractor shall cover and protect all work and materials throughout the course of the work. He shall be responsible for all damage done to, or caused by, this work until the building is completed and accepted by the Owner, and at completion shall leave the work clean.
- B. This Contractor shall cooperate with all other contractors on the project and shall prosecute the work as fast as the progress of other work will allow. He shall consult with the other trades and be responsible for the proper placing and fitting of this work in advance of, or in connection with other work.

1.12 MODIFICATIONS:

- A. Instructions that will modify the work will be ordered by the engineer in writing either before or after the signing of the contract. Only such instructions shall be accepted by this Contractor.
- B. Slight relocations of outlets, devices and equipment shall be made by this Contractor as required, at no additional cost to the Owner, for proper coordination of all trades.

1.13 LOCATIONS AND MEASUREMENTS:

- A. Outlets, devices, fixtures and appliances have been located and shown as accurately as possible on the drawings. The Contractor shall take necessary measurements in the field for exact locations and match his work to the building construction. The outlets for special appliances must be installed so they will be in the proper location when the appliances are installed. Necessary information for the proper installation of these special outlets shall be obtained from the Owner by a request in writing.
- B. This Contractor shall furnish and place all metal sleeves necessary for electrical work at the time of or prior to building of the walls or pouring of slabs. He shall be responsible for the locations and sizes of these sleeves. The Contractor shall furnish and install all inserts and hangers required to support conduits, cables, pull boxes, etc. If the sleeves, hangers, inserts, etc. are improperly installed, this Contractor shall do all necessary cutting and patching at his own expense to rectify the errors.

1.14 EXCAVATION AND BACKFILL:

This Contractor shall do all excavation necessary for the installation of this work and shall backfill such excavation promptly. Backfill shall be thoroughly tamped with mechanical tamper in layers not exceeding six (6) inches. Rocks and rubbish are prohibited from trenches.

1.15 CUTTING, PATCHING, AND REPAIRING:

- A. In new construction, the General Contractor will provide all openings in wall, floor and roof construction required by Electrical Contractor for installation of his work, provided complete information is furnished to the General Contractor at the time required. Failure to provide necessary information will necessitate provisions of additional required openings, chases, recesses, etc., by this Contractor at his own expense, and he shall be fully responsible for the proper cutting and patching of such construction as approved and directed by the engineer.
- B. Where pipes or conduit pass through walls, floors, or roofs, sleeves shall be furnished by this Contractor and installed, except as noted otherwise, by the trade furnishing and installing the material in which they are located. Location of sleeves, inserts, and supports

shall be as directed by this Contractor who will also insure that they are properly installed. Sleeves shall be neatly sawed, sheared, or cut with wheeled cutters. No flame cutting will be permitted.

- C. Each trade shall bear the expense of all cutting, patching, repairing or replacing of the work of other trades required because of his fault, error or tardiness or because of any damage done by him.
- D. Under no circumstances shall the Contractor cut any structural beam or support without prior approval and instructions from the Owner.
- E. If Electrical Contractor installs electrical work through exposed finish walls, ceiling or floor after they are in place, the Electrical Contractor shall close excess openings around his work to match finish surfaces.

PART 2 – PRODUCTS

2.1 WIRING METHODS: Wiring methods shall be:

Rigid Schedule 40 PVC for underground secondary service.

Electrical metallic tubing or MC cable for all systems run above floor inside building.

Rigid Metal Steel Conduit for all systems above floor at Picnic Shelter and exterior.

Liquid-tight flexible conduit for short equipment connections and connections to motors or motor equipment.

Rigid PVC conduit for all systems run under floor or underground.

2.2 INSTALLATION METHODS:

- A. The installation shall comply with the latest issue of all national, state and local rules and ordinances. The requirements of the latest issue of the National Electrical Code shall be considered a minimum. The state and/or local requirements or the plans and/or specifications shall govern when they exceed there requirements of the National Electrical Code.
- B. All work shall be concealed in walls, partitions, ceilings, and floors unless specifically indicated as being exposed. All conduits run overhead shall be run tight to bottom of joists.
- C. In places where walls are masonry, conduits and boxes shall be installed in the walls as they are erected. Boxes shall be of such depth that conduit entrances may be made without excessive cutting of the masonry. Box openings shall be cut neatly to fit the outline of the boxes and the finish plates shall cover the entire cut opening.
- D. Exposed conduit shall be installed with runs parallel or perpendicular to walls and ceilings with right angle turns utilizing outlet boxes or symmetrical bends.
- E. Conduit installed underground shall have a minimum cover of two (2) feet. Joints in conduit installed underground or under floor shall be made watertight. Conduit under concrete floor slab shall be installed on top of subgrade before stone fill is placed.
- F. Exposed conduits shall be securely fastened in place no more than six (6) feet centers for up through one (1) inch and eight (8) feet for all sizes larger than one (1) inch and hangers, supports or fastenings shall be provided at each elbow and at the end of each straight run terminating at a box or cabinet. Horizontal and vertical conduit runs may be supported by one-hole malleable straps, clamp backs, or other approved device with suitable bolts,

expansion shields where needed or beam clamps for mounting to building structure or special brackets. Adjustable hangers may be used to suspend conduits when separately located. Hangers shall be made of durable materials suitable for the application involved.

- G. Conduit ends shall be cut square, threaded and reamed to remove burrs and sharp edges. Field threads shall be of same type and have same effective length as factory cut threads. Conduit joints shall be made with approved couplings. Bends and offsets shall be avoided where possible, but where necessary factory elbows shall be used for one (1) inch and larger. Other offsets and bends shall be made with an approved hickey or conduit bending machine. Conduit deformed or crushed in any way shall not be installed and bends with a radius of less than 3-1/2 inches will not be permitted.
- H. Conduit shall be securely fastened to all sheet metal outlet, junction, cabinets and pull boxes with double galvanized locknuts and insulated bushings, care being observed to see that the full number of threads project through to permit the bushing to be drawn tight against the end of conduit, after the locknuts shall have been made up sufficiently tight to draw them into firm electrical contact with the box. Insulated bushings of fiber or plastic shall be used on one (1) inch conduit and larger.
- I. During installation, conduit ends shall be capped or plugged to prevent the entrance of foreign matter. The Contractor shall exercise necessary precautions to prevent accumulation of water, dirt or concrete in the conduits during execution of the work. Conduits in which water or other foreign materials have been permitted to accumulate shall be cleaned thoroughly or the conduit run replaced where such accumulation cannot be removed by methods approved by the engineer.
- J. No wires shall be installed until work which might cause damage to the wires or conduits has been completed. Conductors terminating at wired outlets shall extend at least eight (8) inches beyond the outlet to facilitate installation of wiring devices or fixtures.
- K. All taps, splices, and joints for conductors size number eight (8) and larger shall be made by mechanical means. Connectors used shall have Underwriter's Laboratories approval for use at six hundred (600) volts. All conductors size number ten (10) and smaller shall be jointed by non-corrosive flux solder connections using the hot dip method except that ideal "wirenuts" or approved equal, as for example, 3M preinsulated (plastic covered) "Scotchlox", shall be used to connect fixture lead wires to their designated branch circuit in junction boxes. These splices shall be electrically and mechanically secure and installed equal to or exceeding the conductor capacity in each instance. All joints, splices, and taps and other sections or wiring requiring taping shall be taped with at least two (2) layers of approved gum rubber tape which shall be laid on with half lap followed by at least one (1) layer of friction of or plastic tape laid on with half lap. The intent of this specification is that the taping shall be neatly done and form a permanently secured insulation equal to that of the wire.
- L. Minimum wire size for all branch circuits shall be No. 12 AWG.
- M. Minimum size for conduit shall be 3/4 inch.
- N. No wiring will be installed in the telephone conduit. Provide pull wires in each conduit.
- O. Where more than one (1) device is installed at the same location and mounting height, a gang plate shall be used.
- P. Junction and/or pull boxes shall be furnished and installed where necessary to avoid excessive runs or too many bends between outlets.
- Q. Powdered soapstone or approved lubricants shall be used to facilitate pulling conductors into conduit.

- R. The following color code shall be used for all conductors in feeder and branch circuits.

120/208, 3 phase

Phase	Black
Phase	Red
Phase	Blue
Neutral	White
Ground	Green

Branch circuit conductors shall be factory color coded. Scotch tape of the proper color may be used to identify conductors No. 4 and larger. All feeders, sub-feeders to panels, motor starters and motors shall be completely phased out as to sequences left to right when facing equipment.

- S. Conductors shall be stranded except that sizes No. 10 and smaller may be solid.
- T. All motors with conduit connections shall be connected to conduit system with short length (not more than twenty-four (24) inches of flexible liquid-tight conduit.)
- U. Circuits are shown schematically. Installation shall be as shown on drawings except by written permission of the engineer.
- V. Conductors shall be continuous from outlet to outlet, and no splices shall be made except within outlet or junction boxes.
- W. Conductors for branch circuits whose length from panel to center of load exceeds fifty feet shall be No. 10 AWG unless a larger size is shown on drawings.
- X. All receptacles to be grounded to the raceway system with a No.12 green jumper connected to the bonding screw. Steel City Style G, grounding clips may be used on stamped steel boxes. Bonding jumper shall be installed as the device box is installed.
- Y. A code gauge, green grounding conductor, (considered a non-current-carrying conductor), shall be pulled through the entire raceway system.
- Z. PVC conduit shall be UL labeled, shall be rated for 90 degrees C. wiring, and shall be assembled with solvent weld fittings. Bends shall be accomplished with equipment specifically intended for the purpose, and flame-type heating devices shall not be used. Any portion of the conduit, which has been scorched or deformed because of excessive heat, shall be rejected. PVC conduits shall not be run exposed in any locations.

2.3 MATERIALS AND EQUIPMENT:

- A. All materials used in this work shall be new and listed by the Underwriter's Laboratories, Inc., in every case where a standard has been established for the particular type of the material to be installed. Materials and equipment furnished under this specification shall be the standard products of manufacturers regularly engaged in the manufacture of such equipment and shall be identical.
- B. Electrical Metallic Tubing/Rigid steel conduit: General Electric Electro Galvanized Zinc-Coating.
- C. EMT Fittings: Thomas & Betts - set screw type.
- D. Rigid Fittings: Thread type.
- E. Flexible Conduit: Liquid-tight flexible conduit shall be "Sealtight" type UA as manufactured by American Brass.

- F. PVC Conduit: Type 40, heavy wall rigid.
- G. Insulated Bushings: Rigid conduit - 0-Z Type "A". EMT - Thomas & Betts - "Blue Throat" plastic insulated series.
- H. Insulated Grounding Bushing: 0-Z Type "B".
- I. Pull and Junction Boxes: Pull boxes shall be fabricated from galvanized sheet steel, not less than 16 gauge thick, with covers held in place by corrosion-resistant machine screws. Boxes shall be furnished and installed where indicated on the plans or where necessary to facilitate cable pulling and splicing. Box size shall be as required by N.E.C. for the number of conduits and conductors entering and leaving it. Where feeder splices are to be made, box shall be large enough to provide ample work space. Exposed boxes shall not have extra knockouts.
- J. Outlet Boxes: All outlet boxes shall be galvanized steel, at least 2-1/8" deep and of sufficient size to accommodate devices noted. Boxes for fixtures shall have fixture studs in the center. If flexible connections are to be made, use "Greenfield" with appropriate fittings for ground continuity. Outlet boxes, for wiring devices in finished walls, shall be one-piece, standard, gang-type, of a size to accommodate number of devices noted. Boxes shall have tile rings to bring box openings flush with finished wall or not more than 1/16" back of same. Sheet steel boxes shall be as manufactured by Steel City, Raco, Appleton, Kilark, or approved equal. Exposed boxes shall not have extra knockouts.
- K. Wiring devices shall be as follows:

Unit	Catalog No.	Capacity
One pole switch	See plans	20A 120/277V
Three way switch	See plans	20A 120/277V
Four way switch	See plans	20A 125V
Duplex receptacle	See plans	20A 125V
Ground fault receptacle	See plans	20A 125V

Equivalent wiring devices shall be as manufactured by P&S, Leviton or Eagle.

Wiring device trim plates shall be as noted on the plans.

See plan for additional information.

- L. Wire and Cable: All wiring shall be of copper. Provide the AWG size shown on the drawings with insulation indicated below. Conductors shall conform to latest NEC requirements and meet ASTM specifications. Conductors shall be stranded except that sizes No. 10 and smaller may be solid. Wire size, voltage rating, insulation type and manufacturer shall be clearly marked on the conductor jacket at regular intervals. Conductors with the following types of insulation shall be installed in locations indicated and/or shown on drawings. Conductors and cables shall be as manufactured by Encore Wire Corp., Southwire Company, or General Cable.

Type THHN, or THWN for No. 10 and smaller branch circuits. (Circuits of 600 volts or less.)

Type THHN or THWN for No. 8 and larger. (Circuits of 600 volts or less.)
- M. Wire Connectors: Pressure connectors as manufactured by "O-Z" or Thomas and Betts for feeders; and wire nuts by Ideal Industries or "Scotch Lox" by Minnesota-Mining for branch circuits.

N. Panelboards:

All breakers shall be bolt-on type. Breakers shall be thermal magnetic tripped and shall have interrupting capacities required on the drawings. Panelboard shall have asymmetrical fault current rating equal to or greater than panels specified on drawings.

Panelboards shall be provided with size and number of breakers as indicated on plans. Main breakers, where required, shall be top or bottom mounted. Side mounted main breakers or those occupying space which will limit panelboard construction to less than 42 branch circuit poles are precluded.

Single pole breakers for lighting circuit, only, may be quick lag type, equipped with one operating handle, molded in one common case. External connections with internal trip devices for ganging quick lag breakers will not be permitted. All breakers to be thermal magnetic, temperature compensated type and shall be quick-make, quick-break type for manual and automatic operation. When panels are used for switching, breakers shall be specifically approved by Under-writers' Laboratory for such operation.

Conductors in panelboards shall be grouped together and laced with plastic ties in a neat, substantial and approved manner.

A neatly typed directory, properly identifying each circuit, shall be installed in each panelboard. A temporary running directory shall be maintained during construction.

Branch circuits shall be connected in each panel board as indicated on drawings.

All panels shall be marked with its designation and voltage by installing engraved plastic plates with pop rivets. Pressure sensitive plastic tape will not be permitted. Submit sample for approval.

Panels shall be Square D or equivalent by General Electric, Siemens, or Eaton. Refer to panel diagrams for required features.

O. Disconnects:

Disconnects shall be furnished and installed at locations shown on plans. Refer to schedule on drawings for additional requirements. Disconnects shall be equipped with fusible poles, as indicated, with quick-break operating mechanism. All disconnects shall be equipped with full-cover interlock.

All disconnects shall be provided with provisions for locking of handle in either "on" or "off" position by installation of pad lock as required. All disconnects shall be equipped with positive pressure fuse clips and shall have visible disconnecting blade switches. Disconnects to be furnished with factory-finished paint with appropriate knockouts for conduit connections.

All disconnects shall be Square D, or equivalent by General Electric, Siemens, or Eaton.

Fusetron, or equal, dual-element fuses shall be used for protection of motor driven equipment.

PART 3 - EXECUTION

3.1 ELECTRICAL CONNECTION OF EQUIPMENT:

- A. Electrical Contractor shall make power connections to all owner supplied equipment. Refer to detail on plans for connections to mechanical and plumbing equipment.

- B. Control wiring for Heating and Air Conditioning Systems is by Heating and Air Conditioning Contractor.

3.2 LIGHTING FIXTURES AND LAMPS:

- A. Catalog number and names of manufacturers are used to convey the types, performance and quality of fixtures to be supplied. Fixtures of other manufacturers may be acceptable provided they offer equal or superior performance and quality and are approved as substitutions. Approval of substitutions 10 days prior to Bid date required.
- B. Catalog numbers given are not necessarily the total fixture specifications. The general description, type and number of lamps, pertinent details as to the quality and functional operation of the fixture as well as the catalog numbers are to be considered in determining the appearance, performance and quality intended. Any major discrepancies in any particular fixture specification should be reported to the engineer prior to submitting a proposal.
- C. It is intended that a lighting fixture be provided for every lighting outlet shown. Any omission is in error and shall be brought to the attention of the engineer prior to submitting a proposal; otherwise the fixture of intent selected by the engineer shall be furnished and installed at no additional cost to the contract.
- D. LED Luminaries:
 - a. LED driver manufacturers should have a minimum of five years of experience with the manufacture of LED drivers. All drivers shall have a minimum warranty of five years.
 - b. Unless noted on the plans, fixtures shall be dimmable down to 10% with standard, electronic, low voltage dimmers.
 - c. Minimum color rendering index (CRI) shall be 80. Color temperature and performance shall conform to the parameters established by ENERGY STAR SSL standards (refer to ANSI-C78.377-2008).
 - d. Optical design shall be low glare, 50% cut-off.
 - e. Rated for 50,000 hours at 70% lumen maintenance.
 - f. LED driver shall be high efficiency with a minimum power factor of .90
 - g. 5 year, 100% warranty coverage for the driver, LED module, housing and trim. For the 1st year this shall be a complete parts and labor warranty. The 4th and 5th years shall cover parts only.
- E. Refer to drawings for fixture schedule.

3.3 PAINTING:

- A. Contractor shall paint all iron and steel not having a factory finish or galvanized finish used for support and hanging equipment throughout building. One coat of primer shall be used followed by one coat of oil base paint with colors selected by the Owner.
- B. All painted equipment with damaged areas shall be painted to match original finish.

3.4 CLEAN-UP:

At the completion of all work, this Contractor shall be responsible for cleaning up all rubbish, leaving the system in perfect operating condition. He shall further clean up rubbish daily in such

a manner that the job shall present a neat appearance. Work shall be accomplished to the satisfaction of the engineer.

3.5 MAINTENANCE AND OPERATING MANUALS:

At the completion of this project the Contractor shall furnish the Owner operating and maintenance manuals containing a brief description of each system and its various components. These manuals should be in accordance with section 07181. Instructions must give full details of the operation of all equipment installed, and shall include manufacturer's printed operating and maintenance instructions, detailed data and bulletins covering all material furnished under the contract giving all necessary illustrations and diagrams and a composite schedule of periodic servicing and lubrication requirements and replacement parts.

3.6 AS BUILT DRAWINGS:

Contractor shall keep and maintain in good order a record of any deviations from drawings for any reason. This record shall be made available to the Owner on the date of substantial completion and shall be legible and accurate so as to be transferable to as-built reproducible drawing.

3.7 GUARANTEE:

The Contractor shall deliver the system to the Owner complete in first-class operating condition in every respect and shall guarantee the material and workmanship for a period of one year from the date of acceptance. If during that time any defect should show up due to defective material, negligence, or want of proper care on the part of the Contractor, he shall furnish such new materials as are necessary to repair such defects and place same in working order at his own expense on receipt of notice of such from the Owner.

END OF SECTION 26000